



Modern human genetic diversity

CN: Discussion of blood and blood types; brief mentions of transatlantic slave trade and racist interpretations of (classical) genetic markers

Dr. Laura van Holstein

lavanho@emory.edu

Room 215

Section 1 checklist

If you feel confident explaining to your granny, or your colleagues not taking Human Biology, or your doctor,

- ☐ what evolution, adaptation, plasticity, developmental plasticity, acclimatisation, epigenetics, natural selection, sexual selection, artificial selection, genetic drift, mutation, and gene flow are,
- ☐ how these processes play a role in generating (or removing) genetic variation within and between groups, and therefore affect genetic diversity,
- ☐ how these processes may explain differences in human phenotypes,
- ☐ across which timescales they operate,
- ☐ what a stressor is (with examples),
- ☐ what buffering is,
- ☐ which biological characters are invariant across the human species,
- ☐ which of these are truly unique to us (relative to Neanderthals) and when and why these may have evolved,
- ☐ which of these are shared among hominins and *not* shared with living apes, and when and why these may have evolved,
- ☐ what the major trends in the intellectual history of the field have been,
- ☐ that much of the early history of the field was unscientific in that data was forced into preconceived conclusions that race was a real biological category,
- ☐ what racial typological thinking is, and what patterns need to be evident in geno- and phenotypic diversity to support it,
- ☐ what modern data shows in contrast to these expected patterns,
- ☐ what the two most important ethical considerations are in conducting research on human subjects – whether alive or deceased,
- ☐ and why these are sometimes hard to put into practice...

then you're on the right track! Great – well done!

Seven main takeaways

Patterns (what?) - *What do we see*

1. Many human traits are nonconcordant
2. Most human genetic variation is clinal and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
4. Humans have (surprisingly?) high genetic homogeneity

Processes (why?)

These are often hard to disentangle!

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit neutral models *→ No selection involved*
3. *but...* ...and others are explained by natural selection

why do we see these patterns

Classical markers

Sickle-cell anemia

Pattern

Process

ABO blood groups

Pattern

Process

DNA markers

SNPs and

microsatellites

Pattern

Process

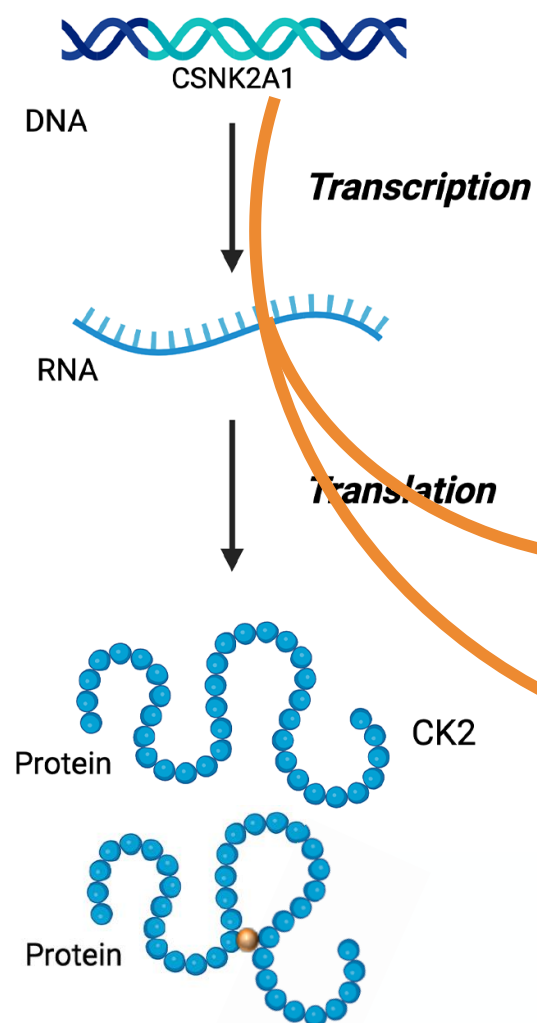
I don't think that anybody uses microsatellites?

What is genetic diversity and **variation**, and **how do these arise?**

- **Variation:** The differences between genomes
 - Encompasses a wide range of scales: from single nucleotide polymorphisms (SNPs) through to structural variation involving millions of base pairs
- **Diversity:** A summary of variation at group level
 - Many measures: heterozygosity, population mutation parameter θ , proportioning of diversity within and between groups (F_{ST})...

Classical versus DNA markers

Central dogma:



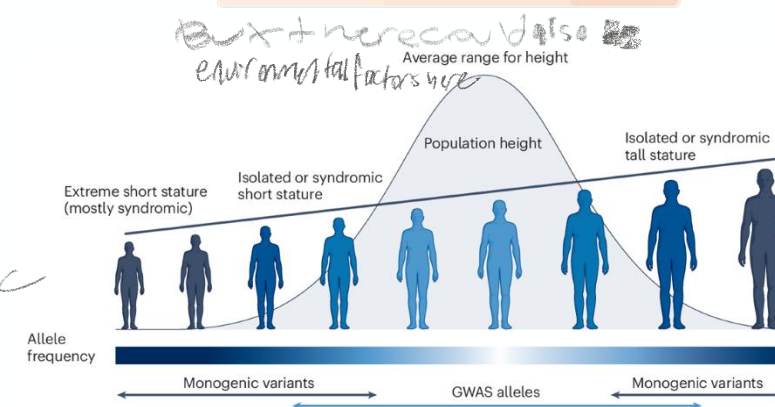
Simple traits

- The trait is either present/absent or in one of two states
- One gene - *monogenic*



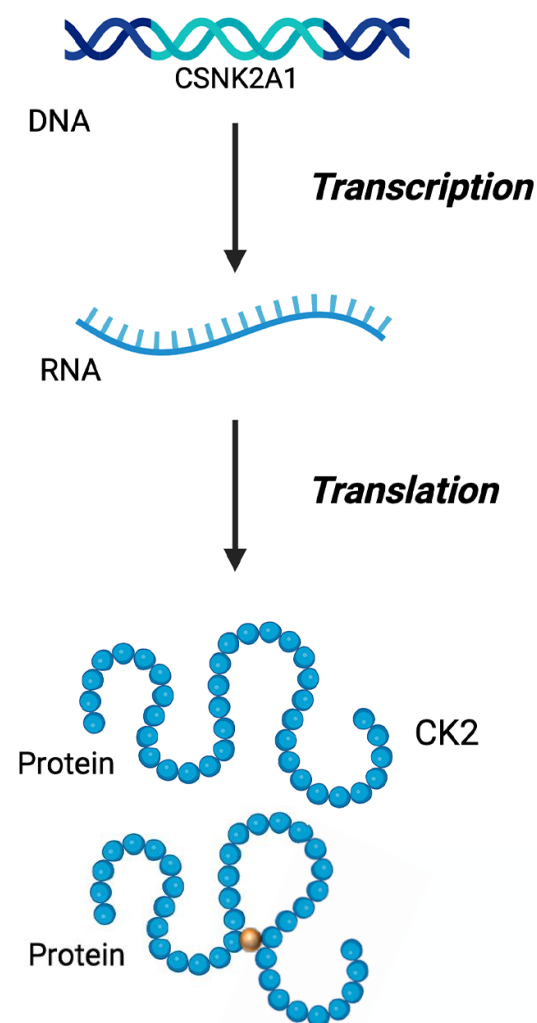
Complex traits

- The trait is continuous
- Many genes - *polygenic*



Classical versus DNA markers

Central dogma:



Classical markers pertain to ...

Only DNA regions directly involved in regions relating to transcription and translation (including genes and sequences that affect their expression).

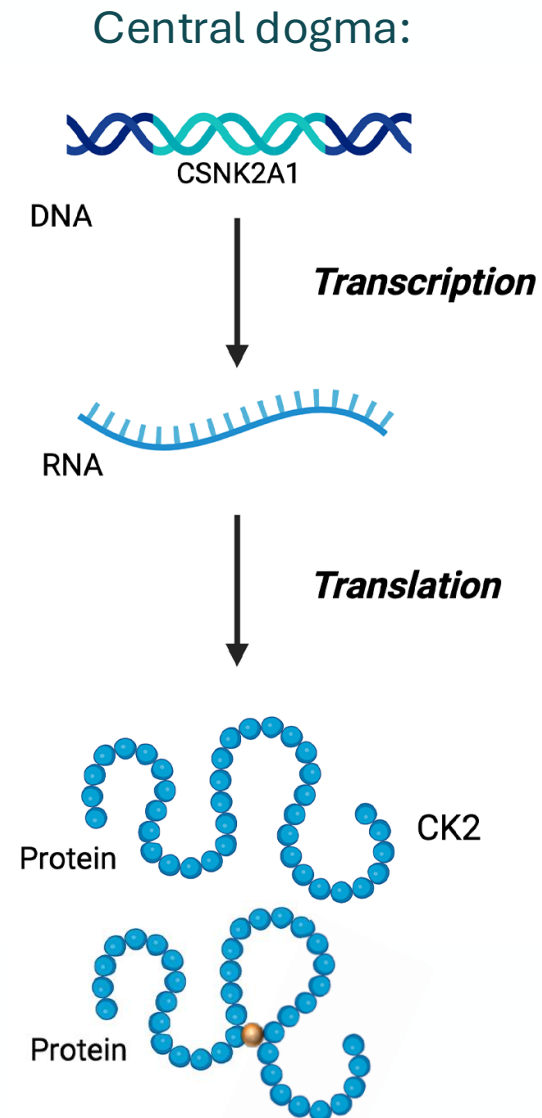
Variation: due to amino acid differences

Classical markers

- Only a very small portion of the genome (~5%)
- Experience natural selection

1. Differences between populations; geographic patterns
2. Adaptive

Classical versus DNA markers



DNA markers

Variation: can derive from anywhere in the genome

- All of the genome
- Most of genome noncoding: no natural selection.
- Reflects other microevolutionary processes: drift, expansions, migrations, admixture.

1. Neutral
2. Clinal patterns that reflect population movement and admixture
3. Coding regions:
 1. Differences between populations; geographic patterns
 2. Adaptive

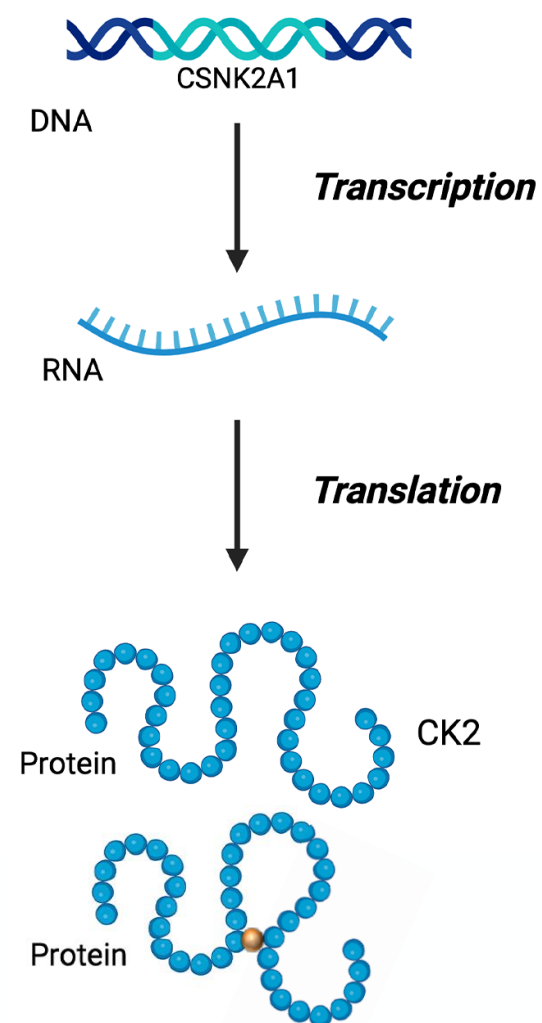
Classical markers

- Only a very small portion of the genome (~5%)
- Experience natural selection

1. Differences between populations; geographic patterns
2. Adaptive

Classical versus DNA markers

Central dogma:



DNA markers

Calafell et al. (2008): O allele (deletion in A allele) has arisen 3 times in human evolution



Classical markers



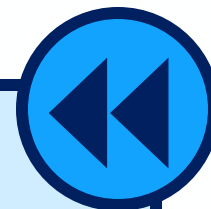
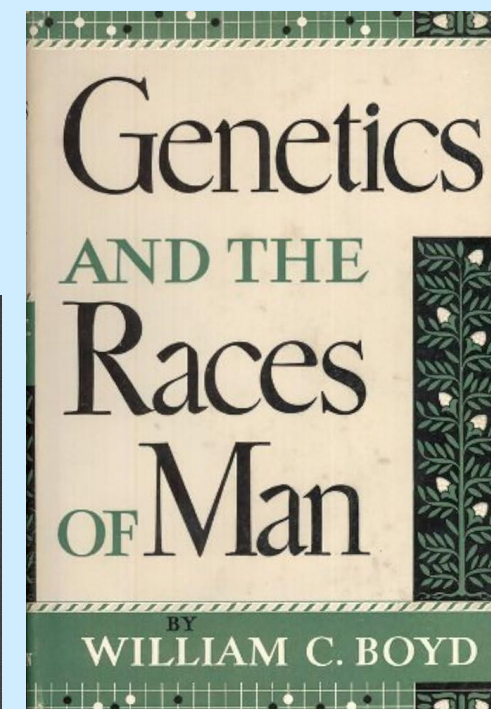
The global distribution of variation in classical markers

- **General geographic patterns**
 - How to interpret these?
- **Group differences (comparing diversity)**
 - Again, how to interpret these?

Our business today is with current, accurate, nonracist descriptions, but...

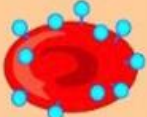
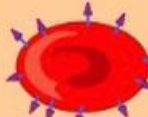
A long and fraught history of interpreting patterns in racist frameworks

➤ See Prof. Lindo's course for more

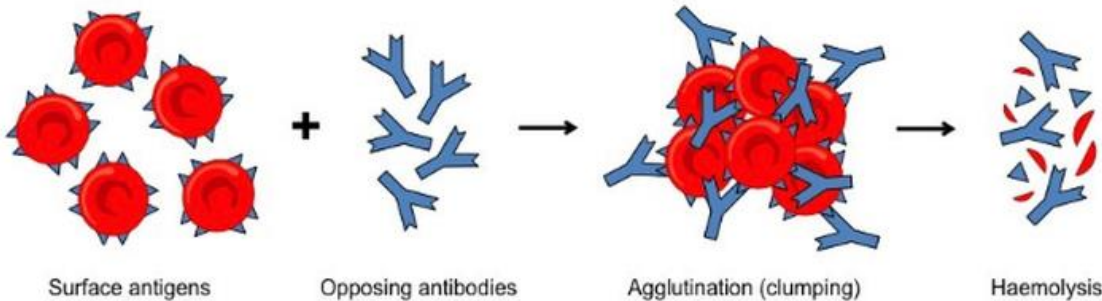


ABO blood groups

Summary of the ABO Blood Groups

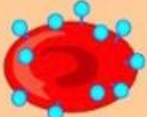
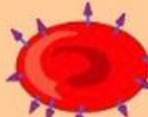
	Type A	Type B	Type AB	Type O
Antigen (on RBC)	Antigen A 	Antigen B 	Antigens A + B 	Neither A or B 
Antibody (in plasma)	Anti-B Antibody 	Anti-A Antibody 	Neither Antibody 	Both Antibodies 
Blood Donors	Cannot have B or AB blood Can have A or O blood	Cannot have A or AB blood Can have B or O blood	Can have any type of blood Is the universal recipient	Can only have O blood Is the universal donor

The Consequence of an Incompatible Blood Transfusion

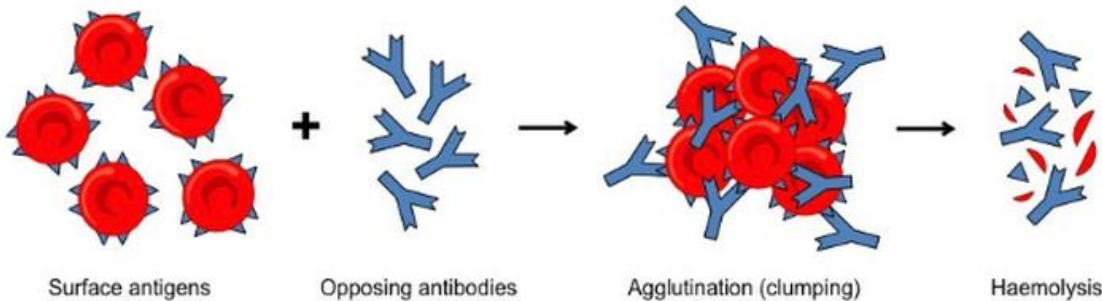


ABO blood groups

Summary of the ABO Blood Groups

	Type A	Type B	Type AB	Type O
Antigen (on RBC)	Antigen A 	Antigen B 	Antigens A + B 	Neither A or B 
Antibody (in plasma)	Anti-B Antibody 	Anti-A Antibody 	Neither Antibody 	Both Antibodies 
Blood Donors	Cannot have B or AB blood Can have A or O blood	Cannot have A or AB blood Can have B or O blood	Can have any type of blood Is the universal recipient	Can only have O blood Is the universal donor

The Consequence of an Incompatible Blood Transfusion



ABO blood groups: global distributions

- 1951; Royal Anthropological Institute (London); Nuffield Blood Group Centre
- Arthur Mourant, a hematologist affiliated with the World Health Organization (WHO)
- Standardized and computed data from Britain's National Blood Transfusion Service, and extracted from published and unpublished results sent in by doctors, transfusion workers, and missionaries around the world.



If interested – relevant for future medics and historians of science: atlasobscura.com/articles/the-quest-to-map-and-archive-human-blood

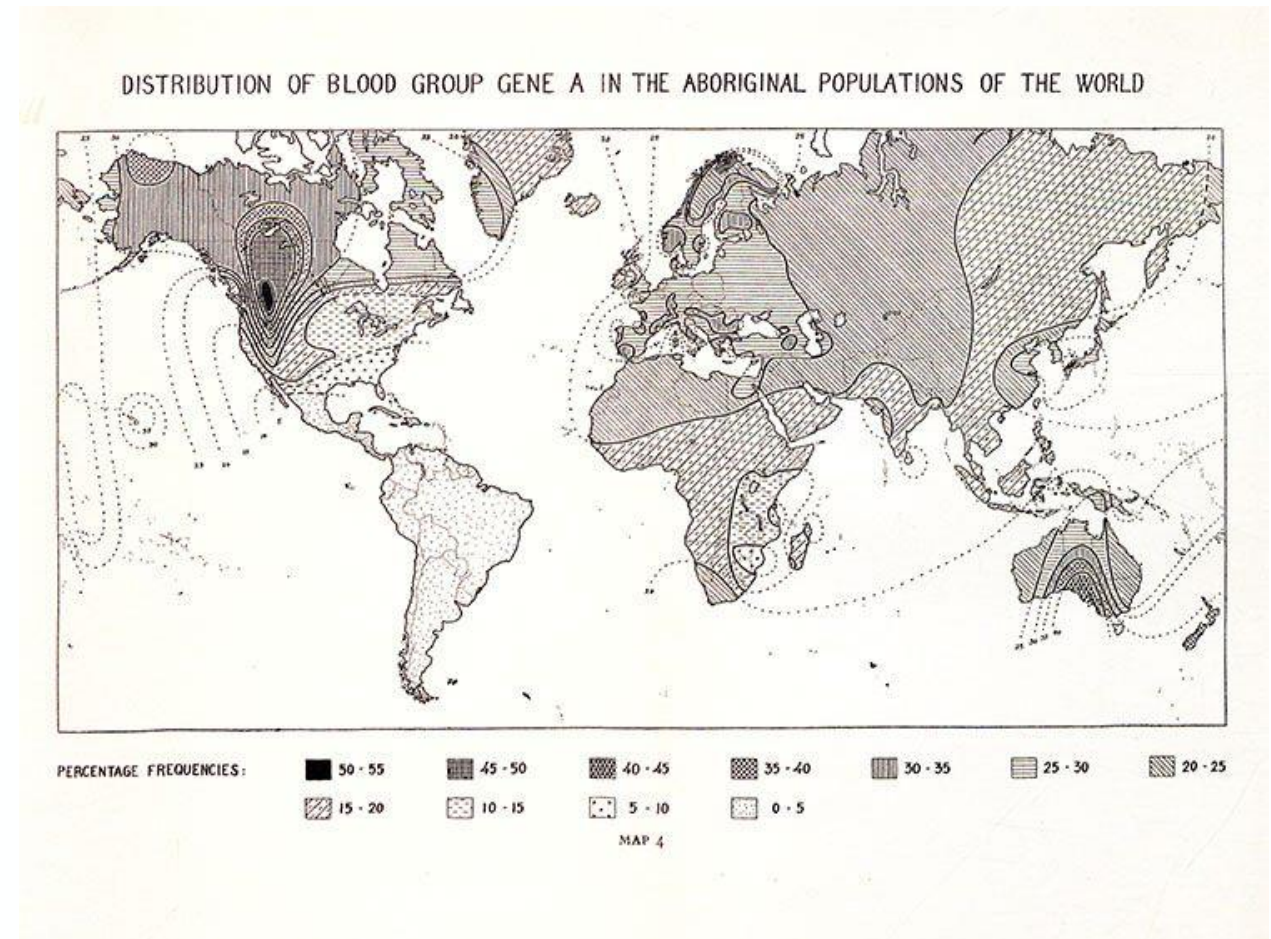
ABO blood groups: global distributions

- 1951; Royal Anthropological Institute (London); Nuffield Blood Group Centre
- Arthur Mourant, a hematologist affiliated with the World Health Organization (WHO)
- Standardized and computed data from Britain's National Blood Transfusion Service, and extracted from published and unpublished results sent in by doctors, transfusion workers, and missionaries around the world.

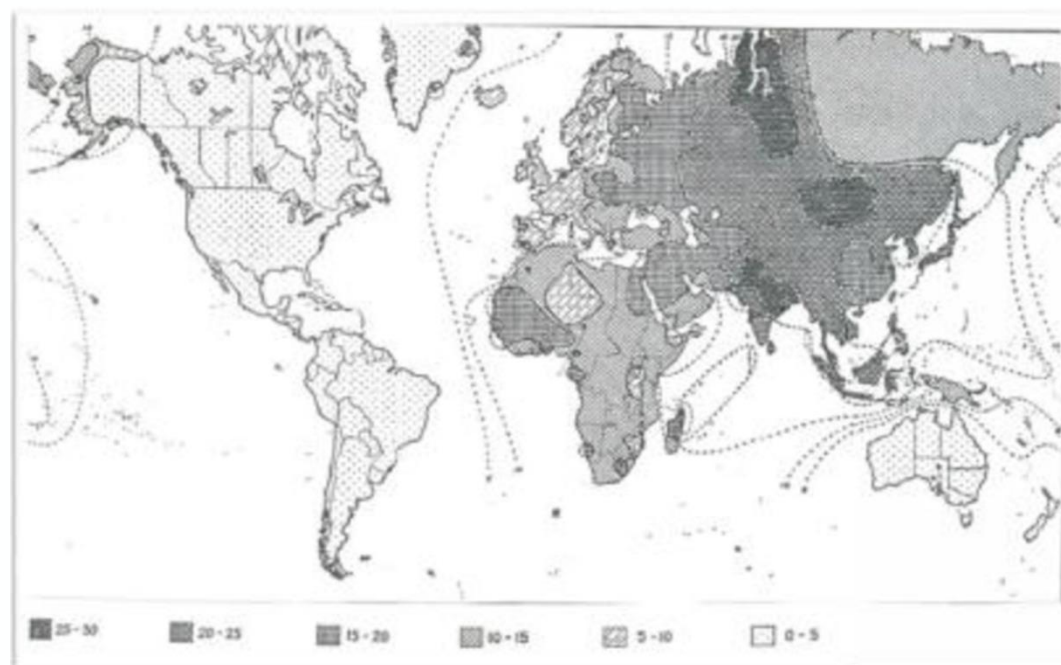


If interested – relevant for future medics and historians of science: atlasobscura.com/articles/the-quest-to-map-and-archive-human-blood

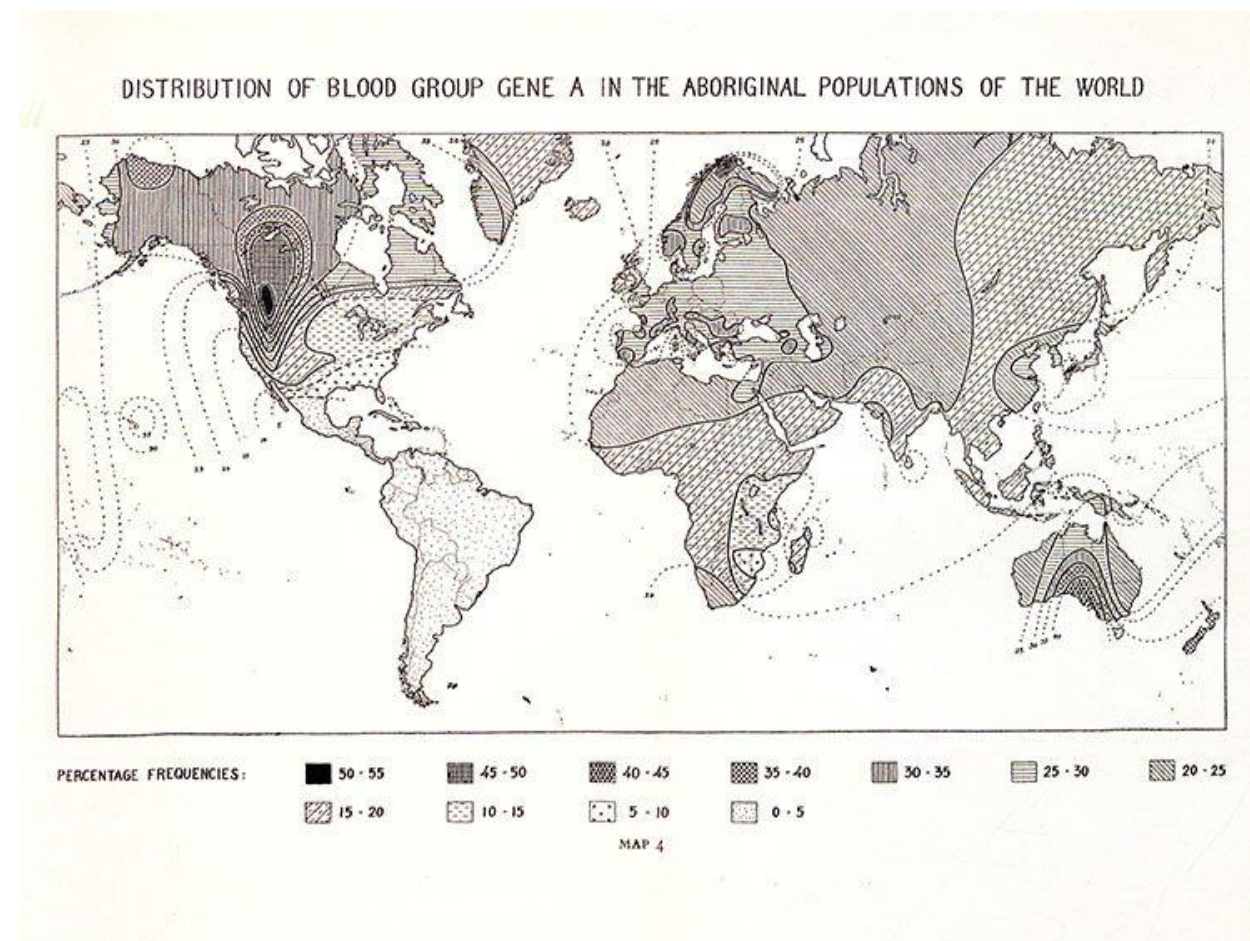
GENOCLINES:



ABO blood groups: global distributions

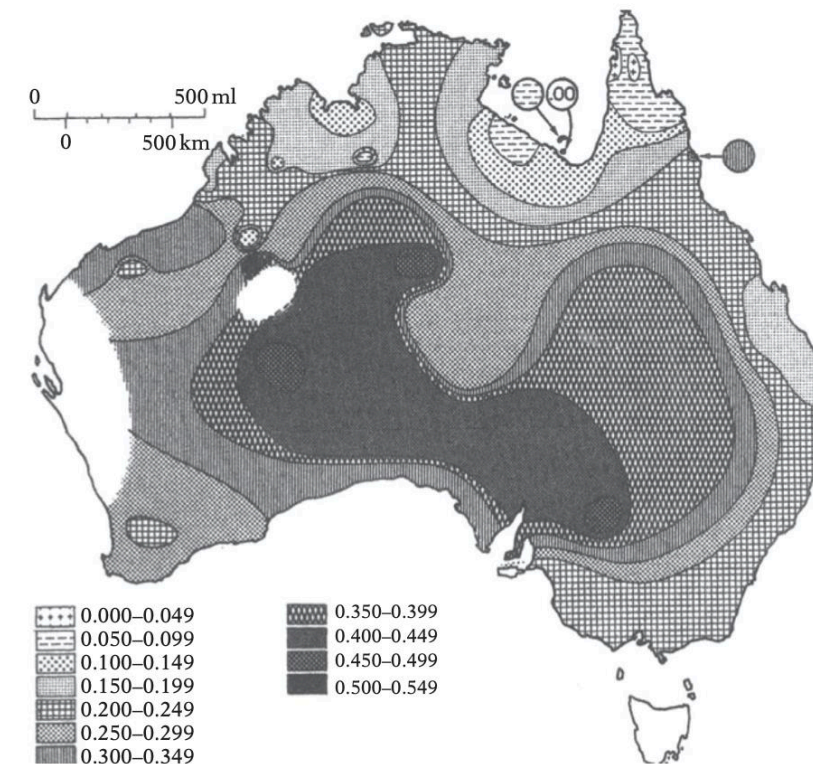
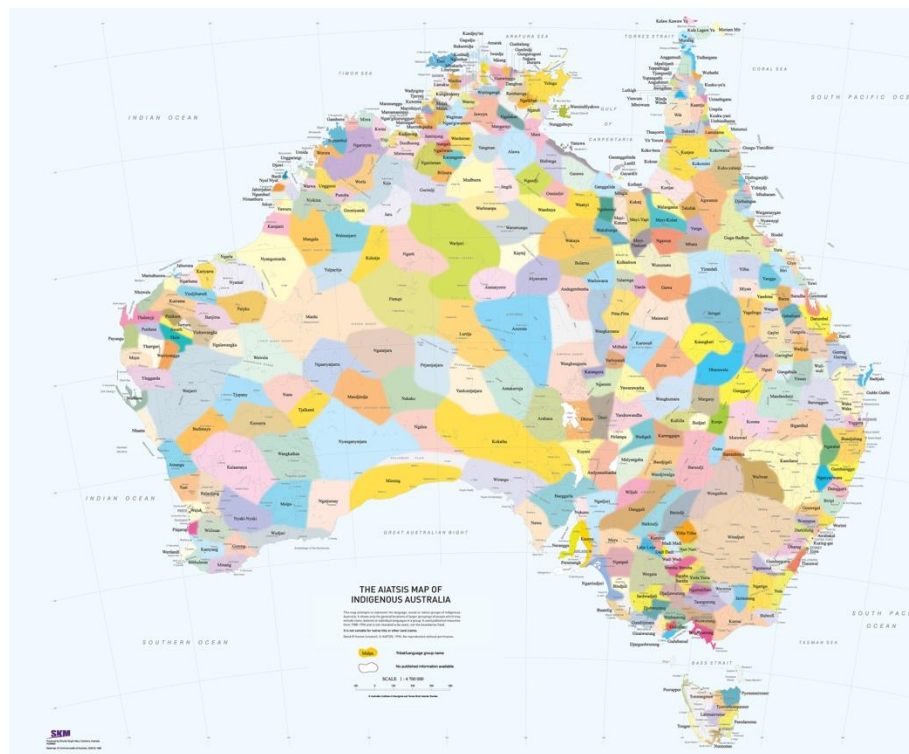


GENOCLINES:

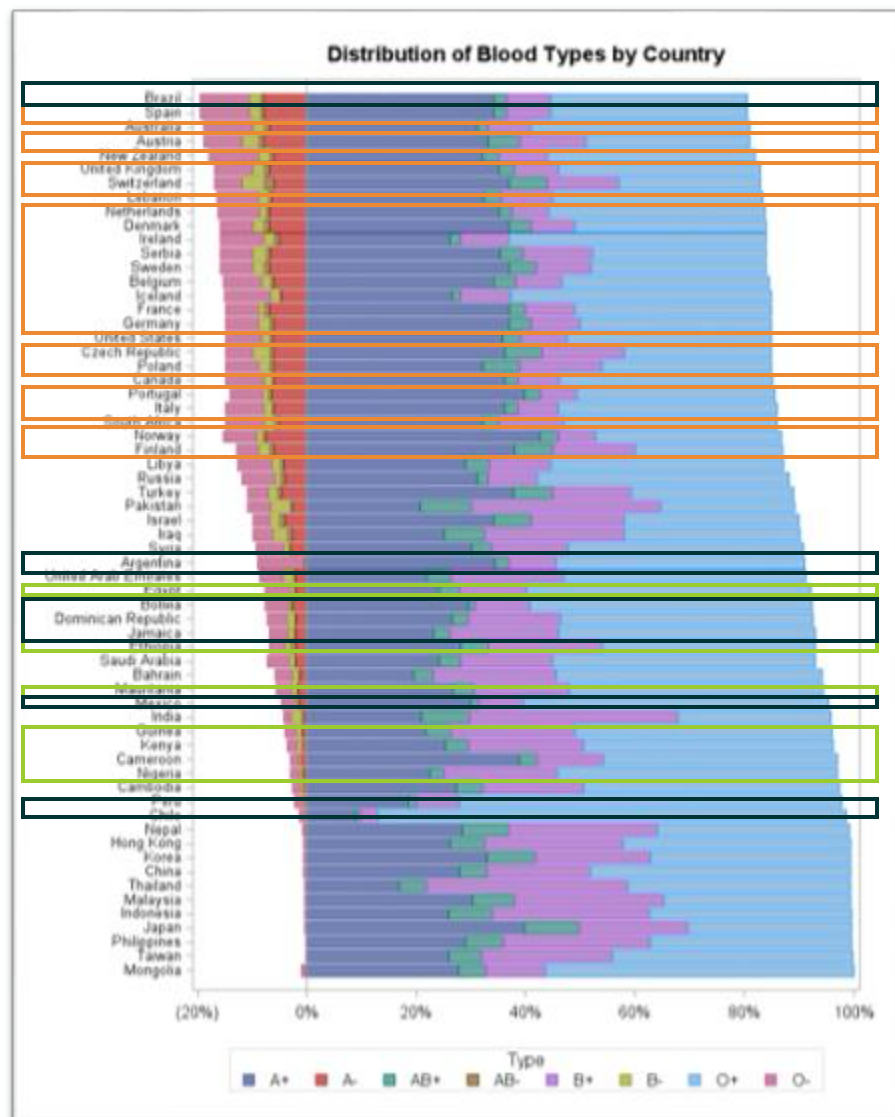


ABO blood groups: continental distributions

- No correlation with modern ethnic groups (linguistic and socially defined)



ABO blood groups: frequencies by country and region



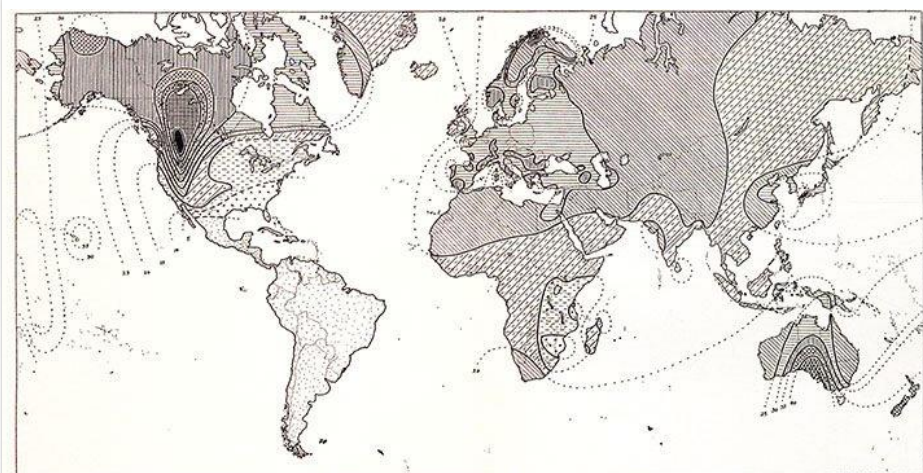
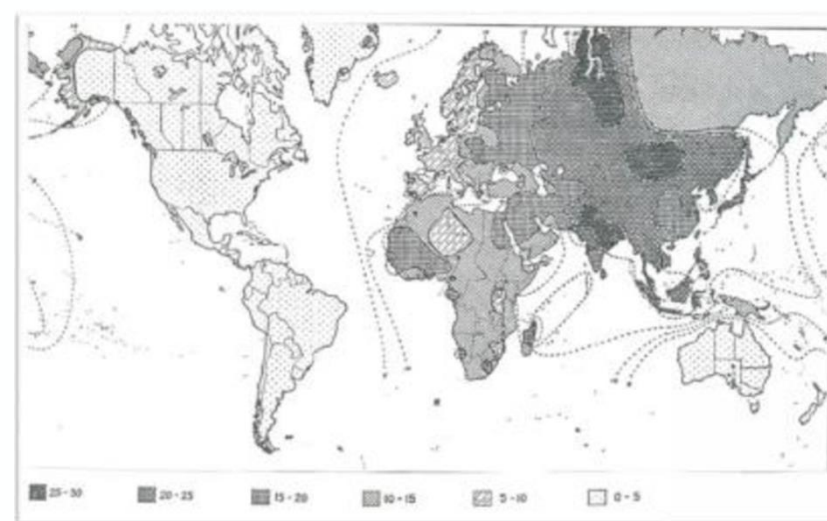
- Europe
- Africa
- South America

Patterns:

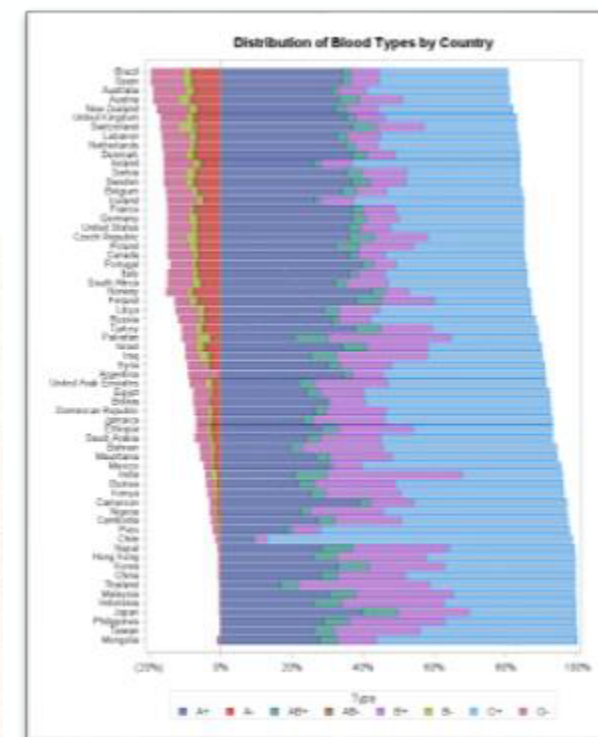
- Some geographic structure
- But no discrete boundaries between regions

ABO blood groups: overall patterns

- No clean breaks between populations
- Blood type frequencies are clinally distributed
- But there *are* geographic patterns
 - Closer together = (somewhat) more likely to have same blood type
 - Frequencies are not the same across the world – some blood types are more common in certain regions



PERCENTAGE FREQUENCIES: 50-55, 45-50, 40-45, 35-40, 30-35, 25-30, 20-25, 15-20, 10-15, 5-10, 0-5



ABO blood groups: overall patterns and their causes

- No clean breaks between populations
- Blood type frequencies are clinally distributed
- But there *are* geographic patterns
 - Closer together = (somewhat) more likely to have same blood type
 - Frequencies are not the same across the world – some blood types are more common in certain regions

1. Isolation by distance

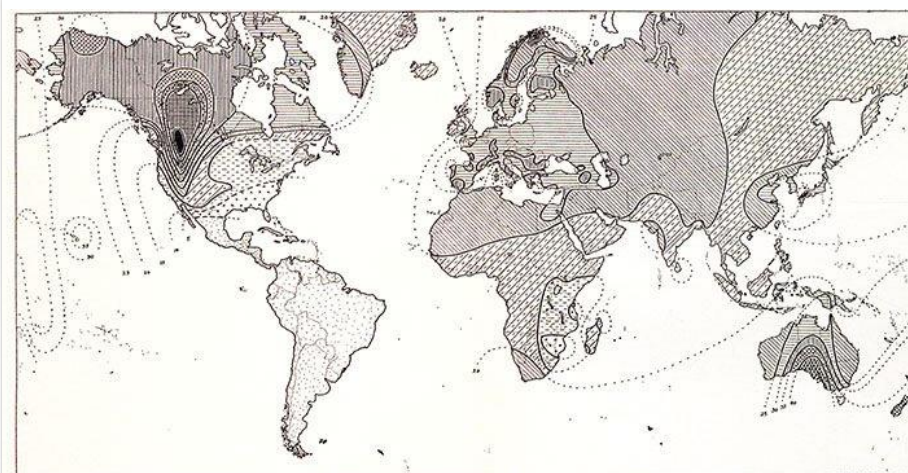
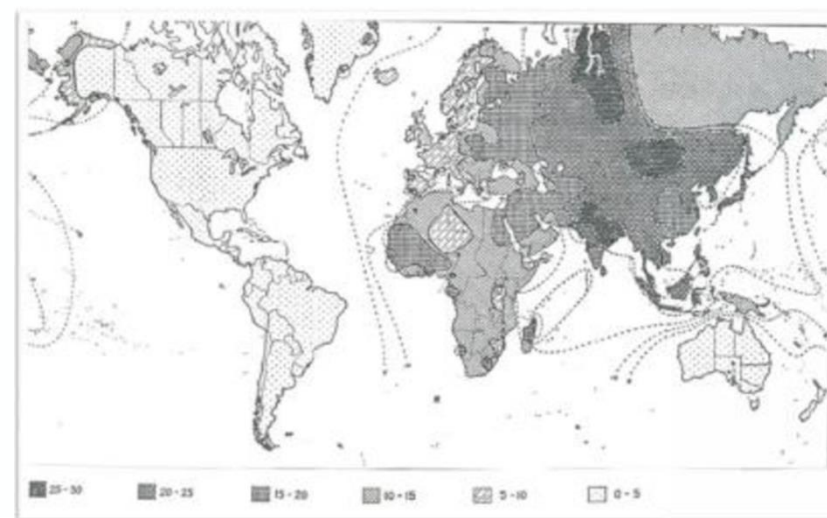
Isolation by distance



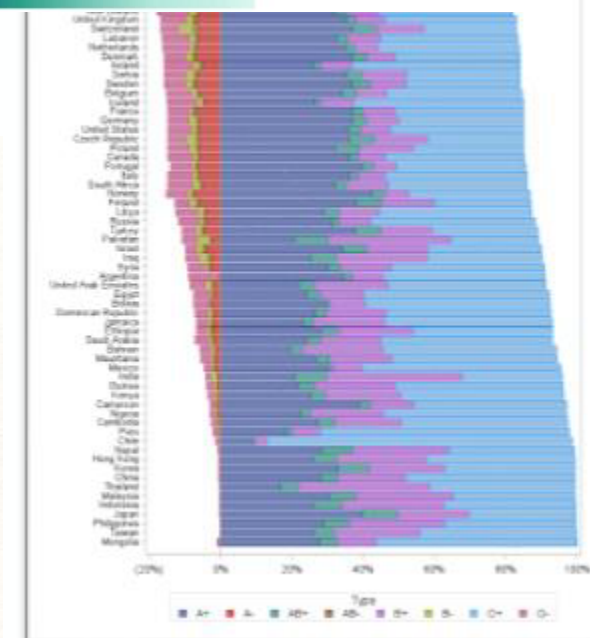
Selection



Genetic similarity



PERCENTAGE FREQUENCIES:

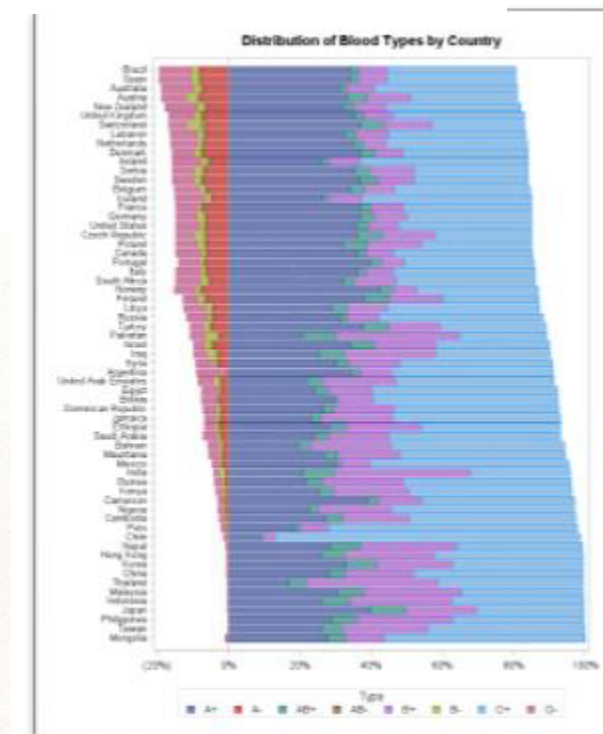
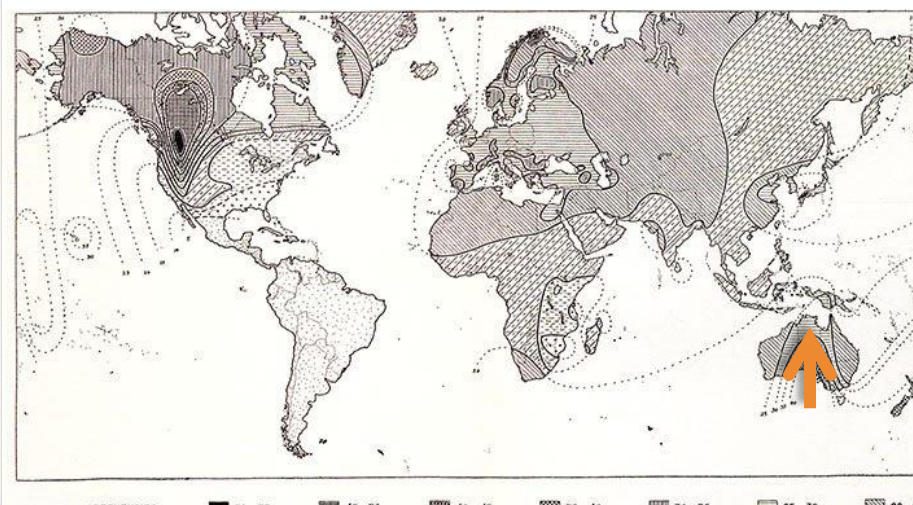
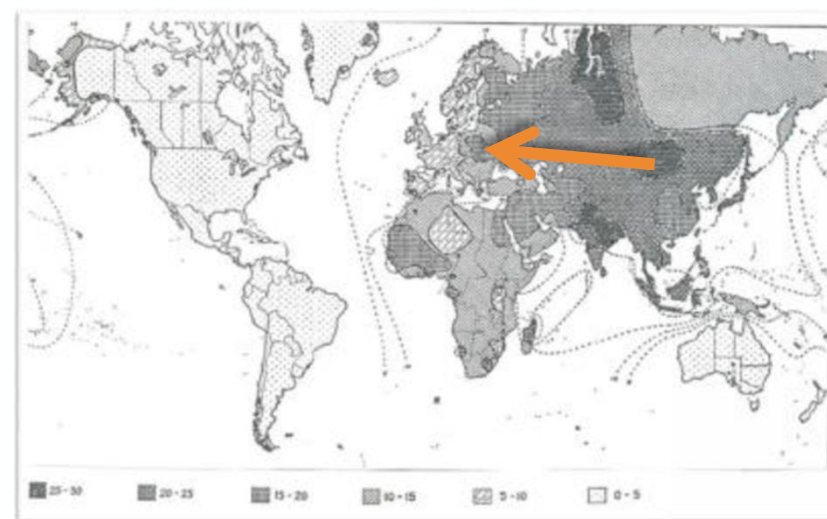


ABO blood groups: overall patterns and their causes

- No clean breaks between populations
- Blood type frequencies are clinally distributed
- But there *are* geographic patterns
 - Closer together = (somewhat) more likely to have same blood type
 - Frequencies are not the same across the world – some blood types are more common in certain regions

2. Migrations

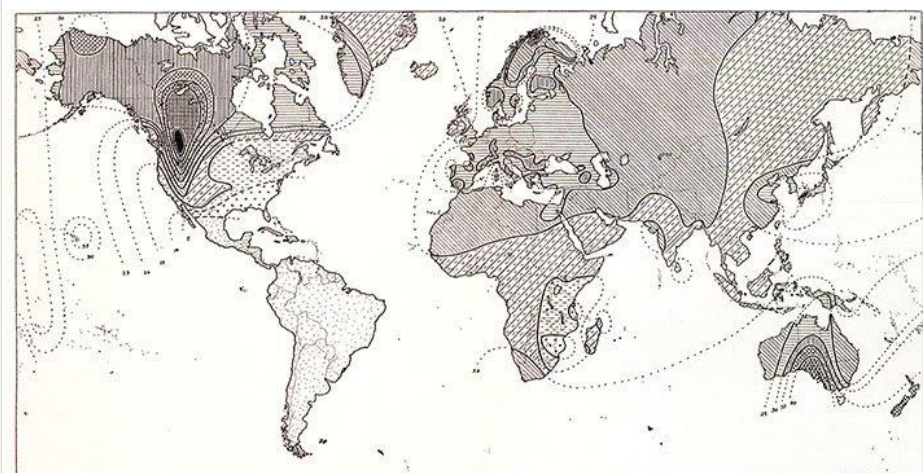
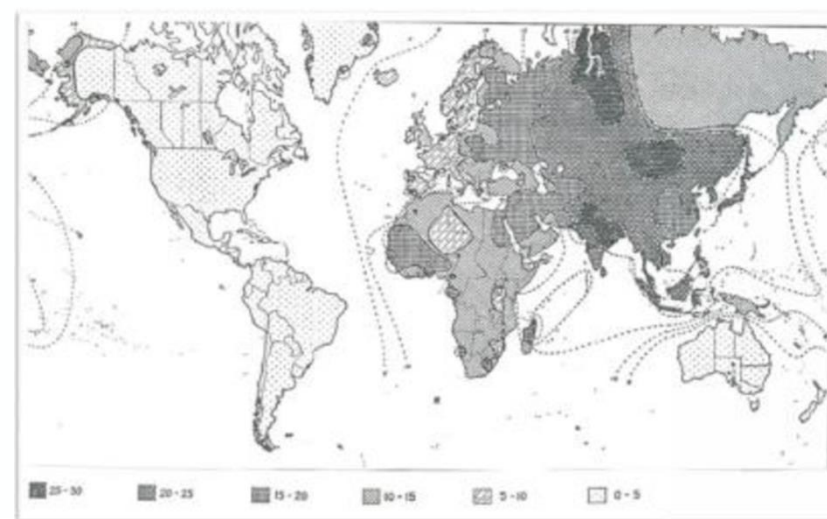
- B: Neolithic farmers expanding from the Middle East from around 11,000 BP, and steppe pastoralists coming out of the Pontic Steppe around 5,000 BP
- A: Successive migration waves from small founder population



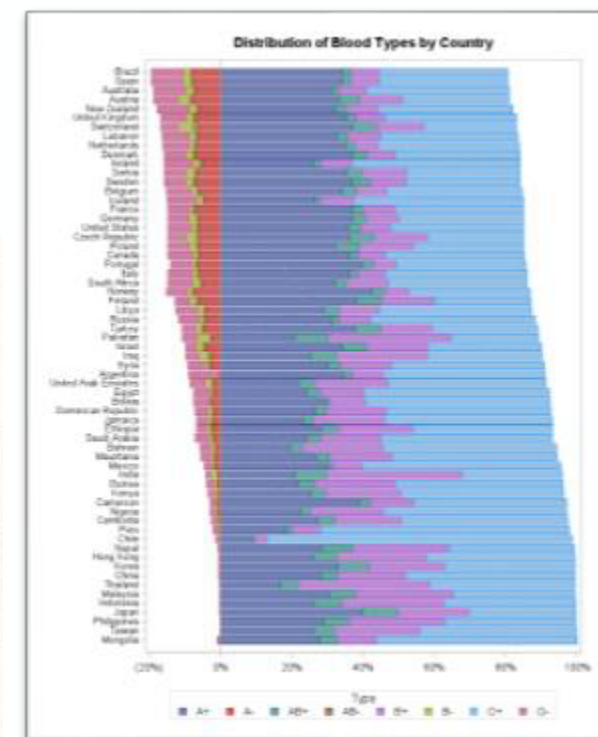
ABO blood groups: overall patterns and their causes

- No clean breaks between populations
- Blood type frequencies are clinally distributed
- But there *are* geographic patterns
 - Closer together = (somewhat) more likely to have same blood type
 - Frequencies are not the same across the world – some blood types are more common in certain regions

3. Selection



PERCENTAGE FREQUENCIES: 50-55, 45-50, 40-45, 35-40, 30-35, 25-30, 20-25, 15-20, 10-15, 5-10, 0-5

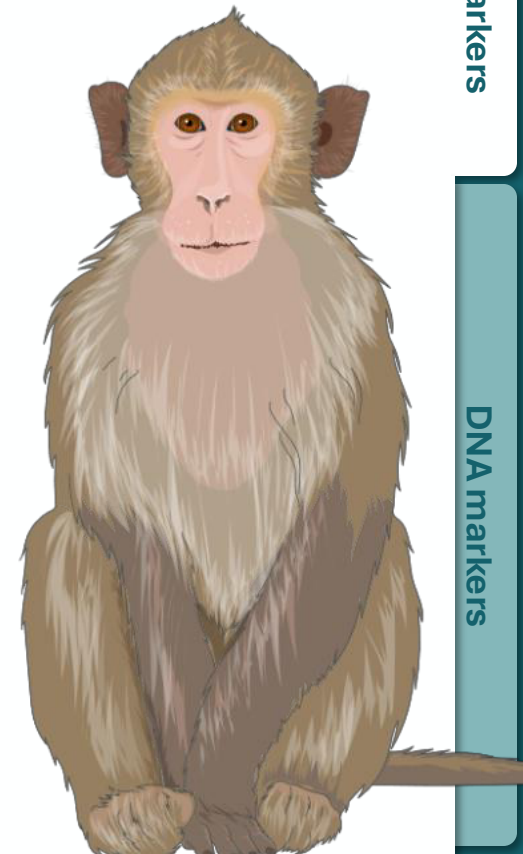


ABO blood groups: overall patterns **and their causes**

- No clean breaks between populations
- Blood type frequencies are clinally distributed
- But there *are* geographic patterns
 - Closer together = (somewhat) more likely to have same blood type
 - Frequencies are not the same across the world – some blood types are more common in certain regions

3. Selection

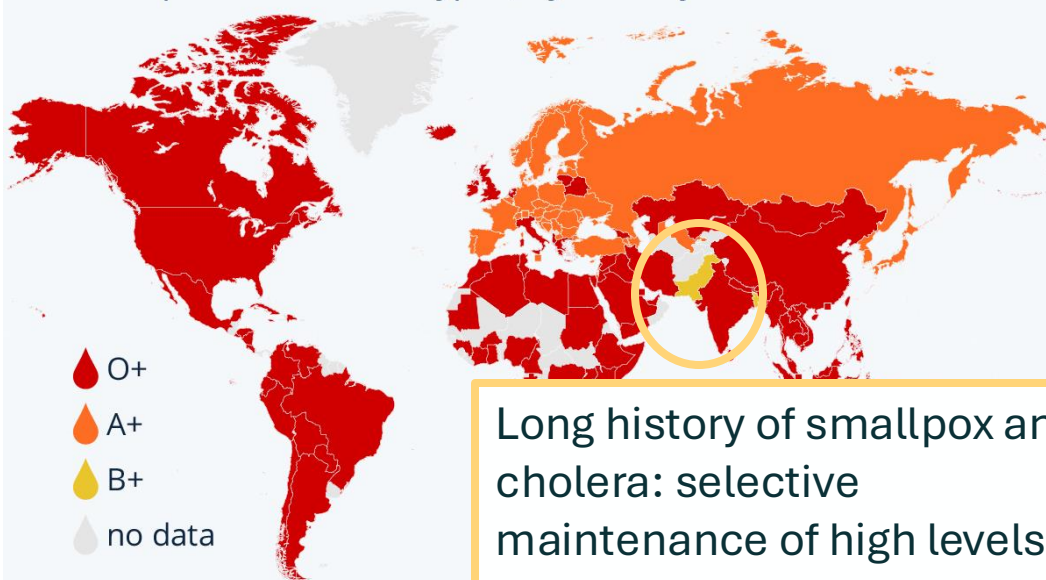
- Rhesus macaques: antigens analogous to A and B.
 - Comparative method suggests polymorphism was maintained throughout human evolution. **Selection?**



ABO blood groups: overall patterns and their causes

The Most Common Blood Types Around the World

Most prevalent blood types, by country



Long history of smallpox and cholera: selective maintenance of high levels of B antigen (Buchi 1968)

Depending on the country, the major blood type varies between 30 percent and 75 percent of the population.
Source: World Population Review

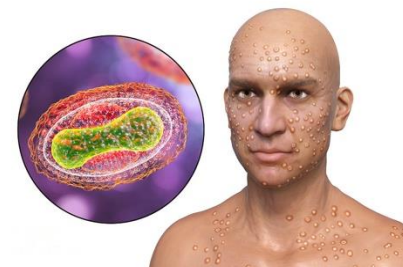


statista

3. Selection

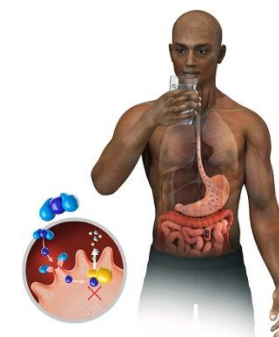
- Rhesus macaques: antigens analogous to A and B.
 - Comparative method suggests polymorphism was maintained throughout human evolution. **Selection?**
- Antigens & antibodies: **point to role for infectious disease**
 - Viral/bacterial antigen similarity to own body's antigen should lead to higher risk of disease
 - But deeper research: varying and inconsistent results
- Non-infectious disease associations strong – but mechanism is unknown**

A antigen (A and AB):
higher risk of **smallpox**



Vogel & Chakravartti (1966)

Neither antigen (O):
higher risk of **cholera**

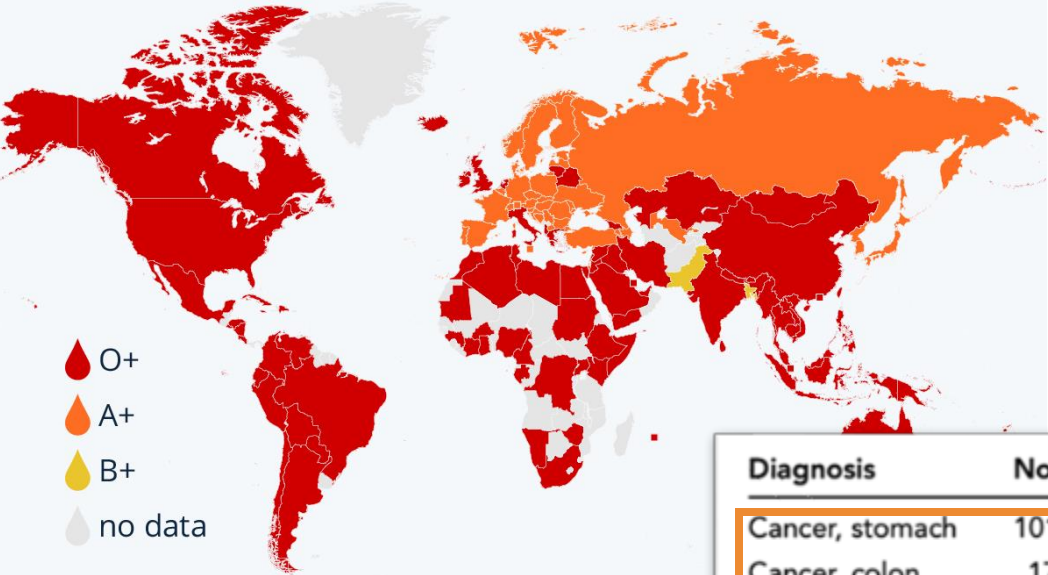


Glass et al. (1985)

ABO blood groups: overall patterns and their causes

The Most Common Blood Types Around the World

Most prevalent blood types, by country



Depending on the country, the majority blood group represents between 30 percent and 75 percent of the population.
Source: World Population Review



3. Selection

- Rhesus macaques: antigens analogous to A and B.
 - Comparative method suggests polymorphism was maintained throughout human evolution. **Selection?**
- Antigens & antibodies: **point to role for infectious disease**
 - Viral/bacterial antigen similarity to own body's antigen should lead to higher risk of disease
 - But deeper research: varying and inconsistent results
- Non-infectious disease associations strong – but mechanism is unknown**

Diagnosis	No. of series	No. of patients	No. of controls	Markers compared	Mean relative incidence*
Cancer, stomach	101	55 434	1 852 288	A:O	1.22
Cancer, colon	17	7 435	183 286	A:O	1.11
Cancer, pancreas	13	817	108 408	A:O	1.24
Duodenal ulcer	44	26 039	407 518	O:A	1.35
Gastric ulcer	41	22 052	448 354	O:A	1.17

Source: Data taken from Vogel and Motulsky (1997, p. 221).
*Relative incidence is determined by the ratio of, for example, A:O in patients compared with A:O in controls.

A greater chance than O
O greater chance than A

Seven main takeaways

Patterns (what?)

1. Many human traits are nonconcordant
2. Most human genetic variation is clinal and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
NO pops w/ own blood type
4. Humans have (surprisingly?) high genetic homogeneity



Processes (why?)

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit neutral models *(high mutation leading to divergence in alleles)*
3. ...and others are explained by natural selection

Seven main takeaways

Patterns (what?)

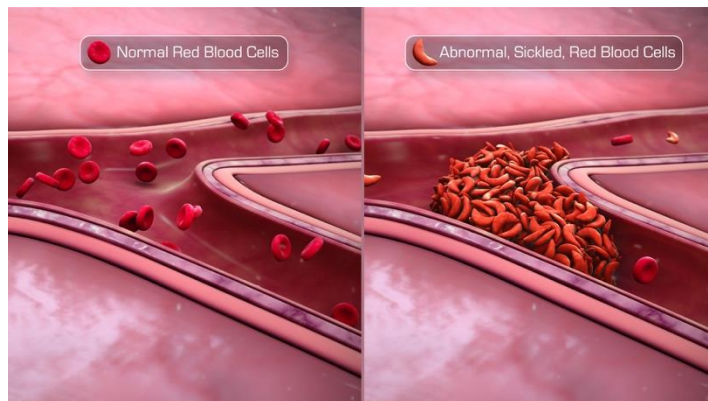
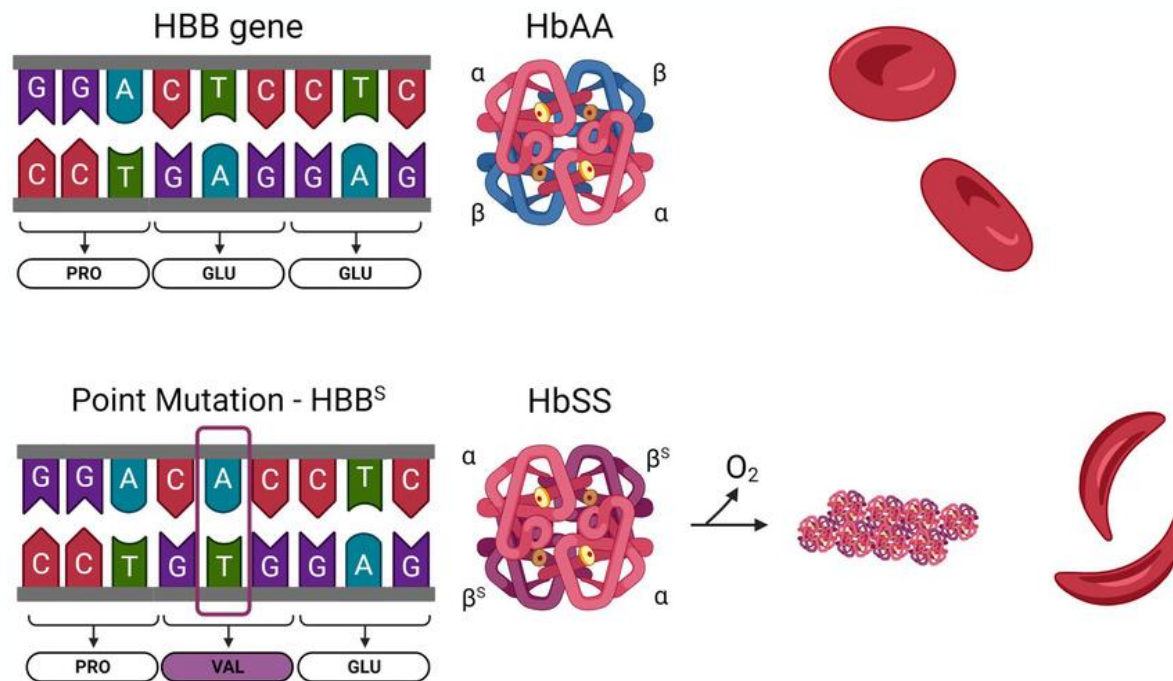
1. Many human traits are nonconcordant
2. Most human genetic variation is **clinal** and/or continuous (not discrete)
3. **Genetic variation is greater within groups than between groups**
4. Humans have (surprisingly?) high genetic homogeneity



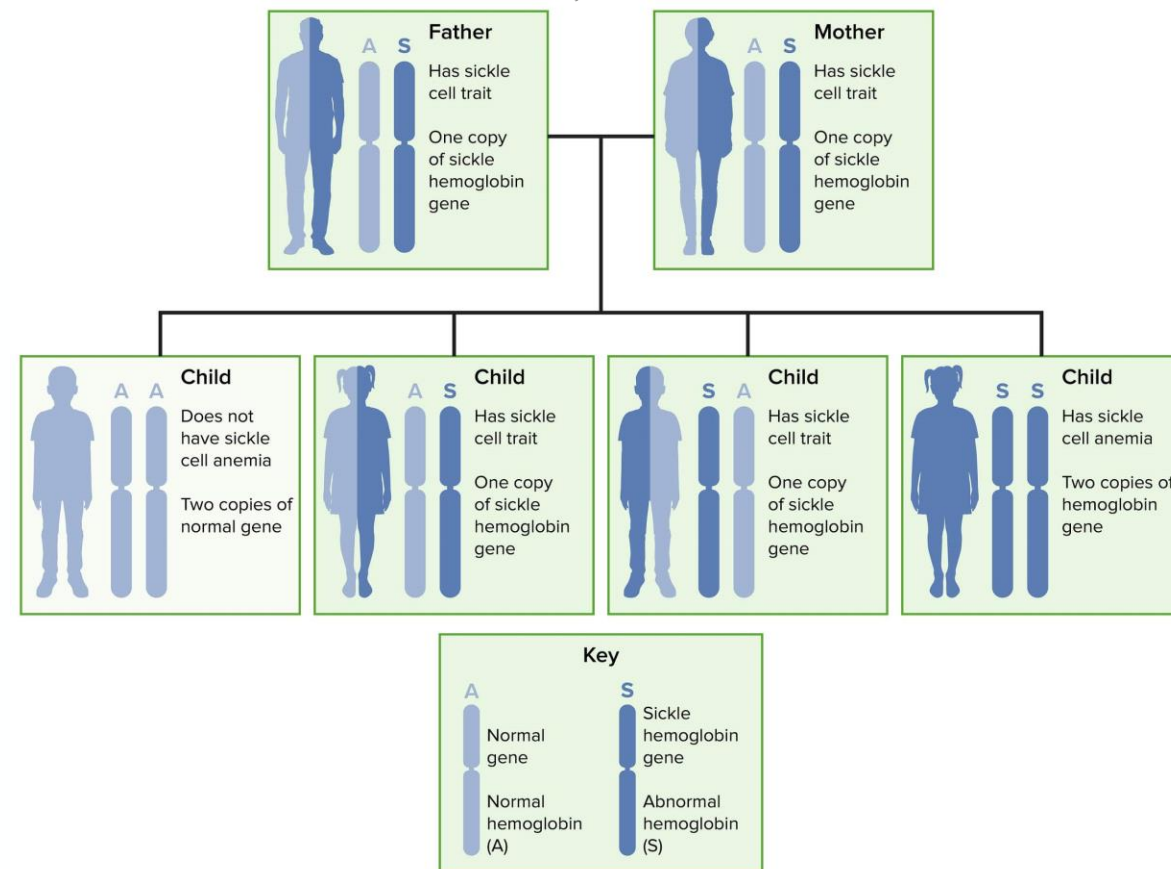
Processes (why?)

1. Genetic data generally fit **isolation-by-distance**, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit **neutral models**
3. ...and others are explained by **natural selection**

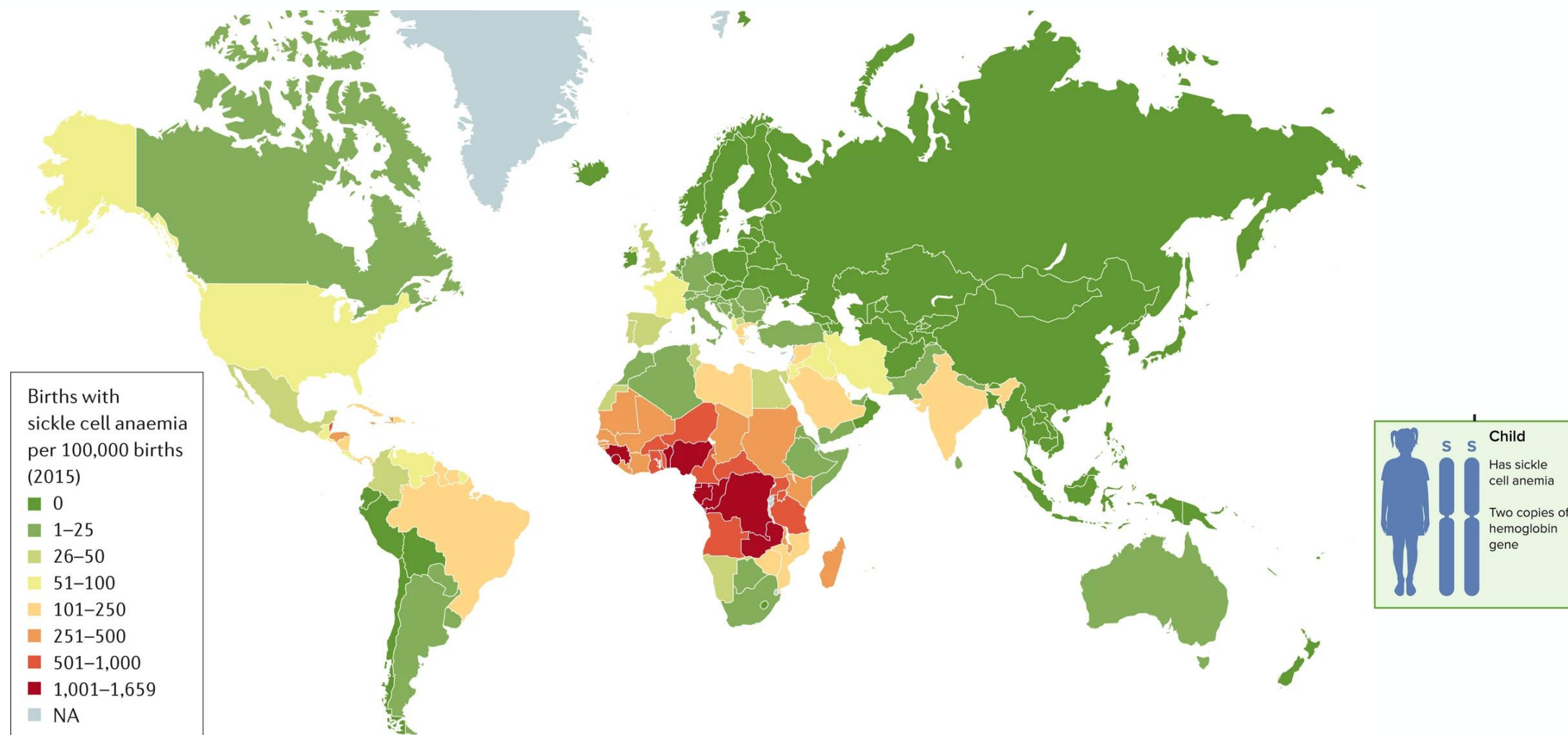
Sickle-cell anemia



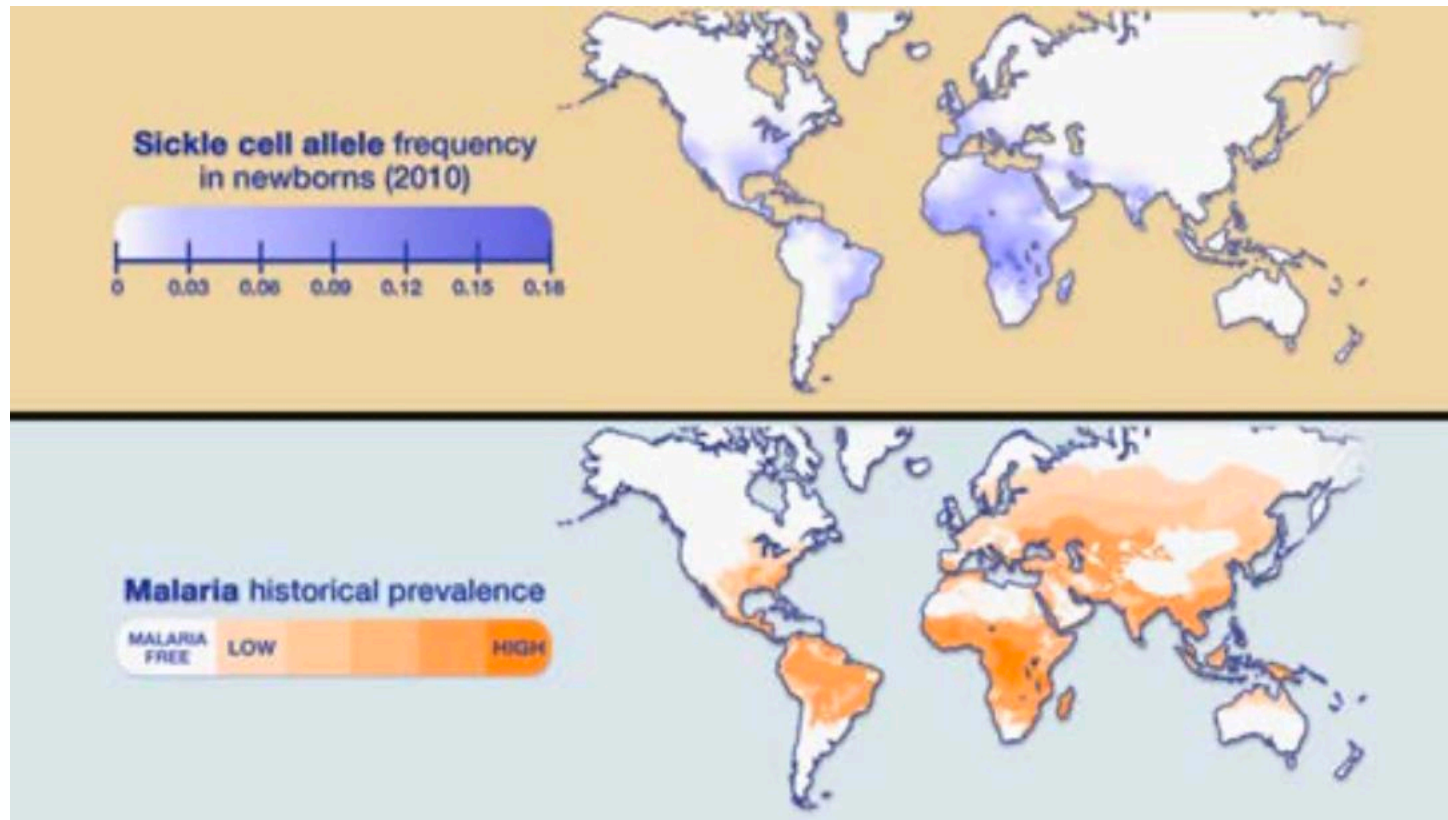
*AA and S are codominant to heterozygosity
 AA and S are both expressed, some normal RBCs
 RBCs some sickle RBCs*



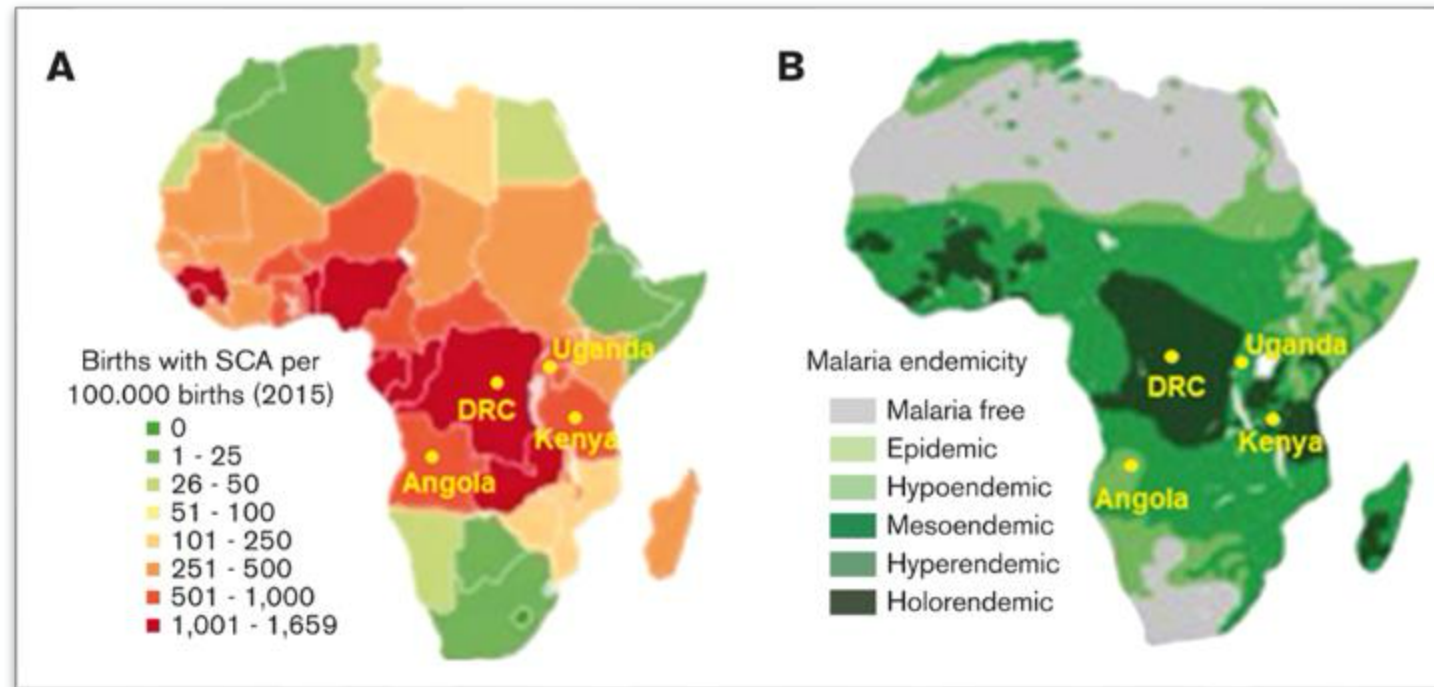
Sickle-cell anemia: a clinal pattern



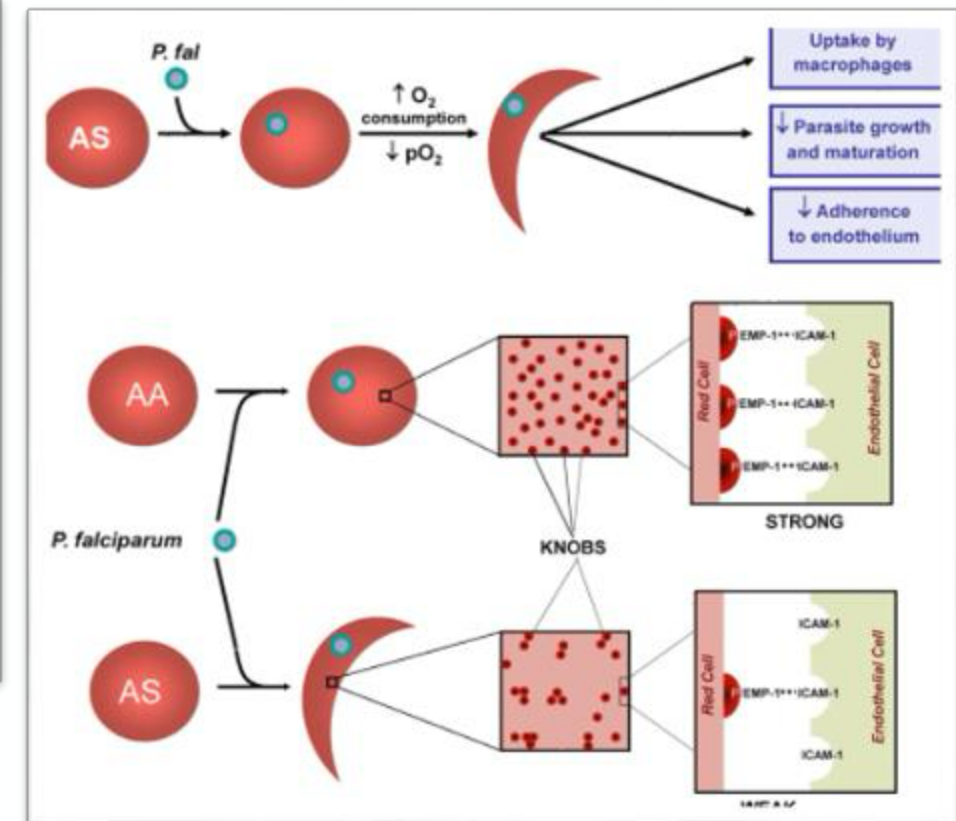
Sickle-cell anemia: a clinal pattern



Sickle-cell anemia: a clinal pattern explained by malaria

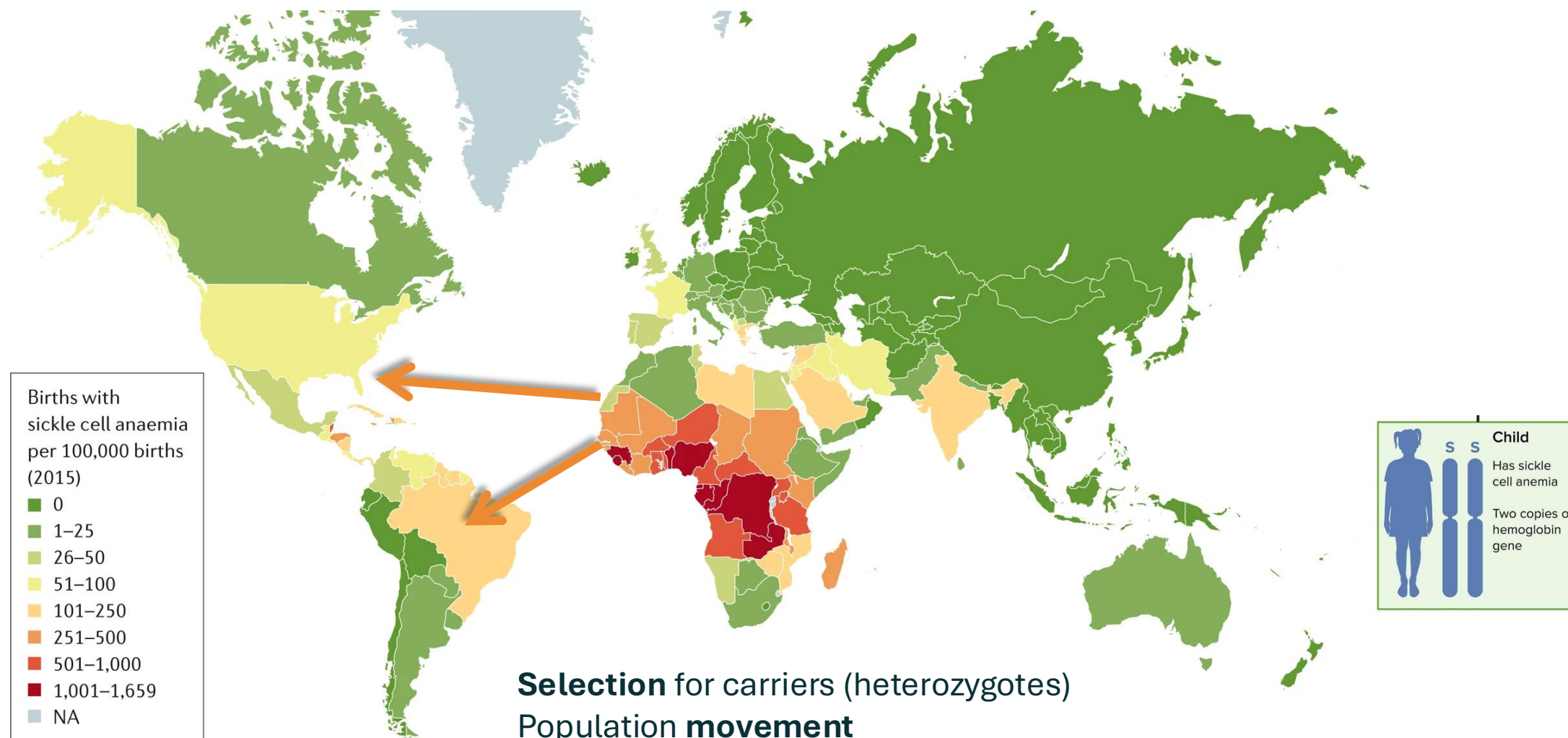


Safeukui et al. (2023)



Bunn (2013)

Sickle-cell anemia: evidence for O-O-A?



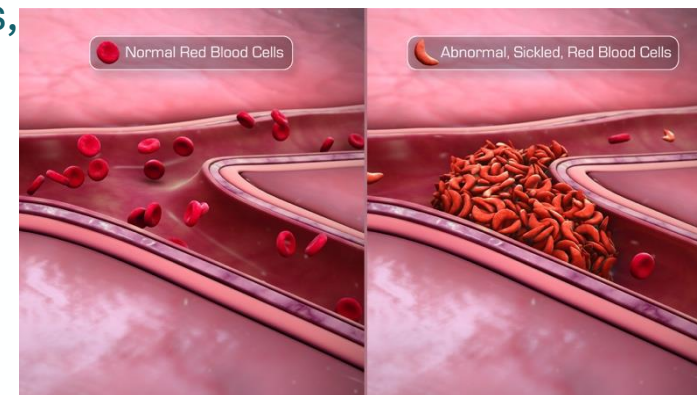
Seven main takeaways

Patterns (what?)

1. Many human traits are nonconcordant
2. Most human genetic variation is clinal and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
4. Humans have (surprisingly?) high genetic homogeneity

Processes (why?)

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit neutral models
3. ...and others are explained by natural selection



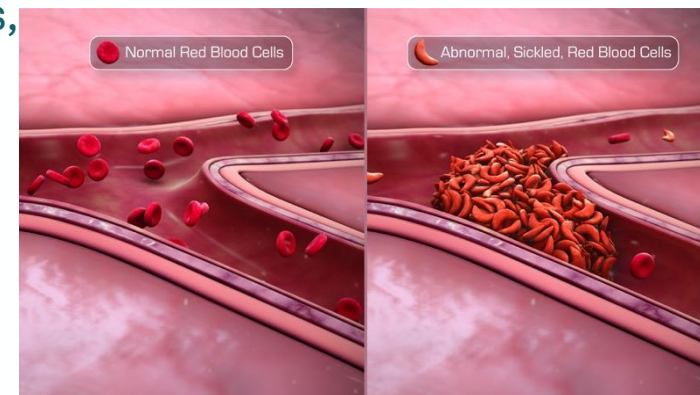
Seven main takeaways

Patterns (what?)

1. Many human traits are nonconcordant
2. Most human genetic variation is **clinal** and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
4. Humans have (surprisingly?) high genetic homogeneity

Processes (why?)

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit **neutral models**
3. ...and others are explained by **natural selection**



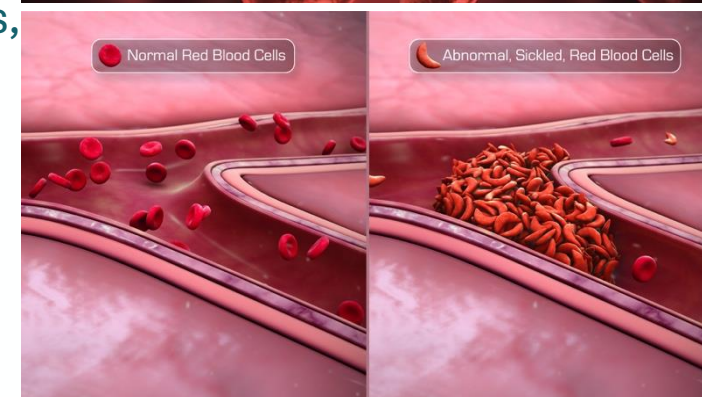
Seven main takeaways

Patterns (what?)

1. Many human traits are **nonconcordant**
2. Most human genetic variation is **clinal** and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
4. Humans have (surprisingly?) high genetic homogeneity

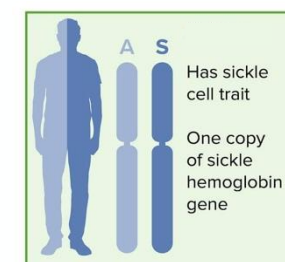
Processes (why?)

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit **neutral models**
3. ...and others are explained by **natural selection**

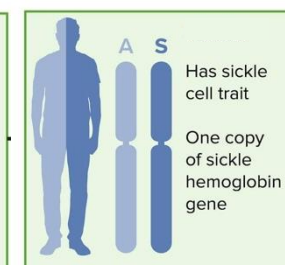
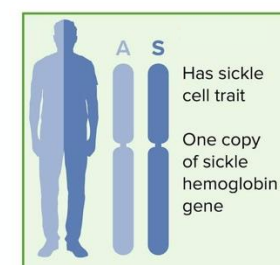
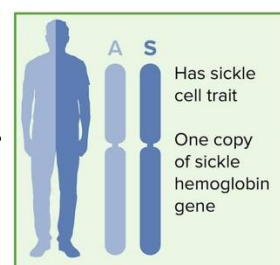
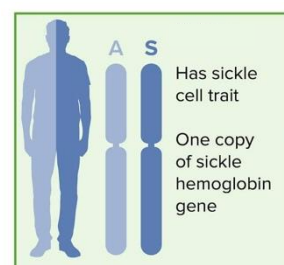


What is genetic diversity and **variation**, and **how do these arise?**

- **Variation:** The differences between genomes
 - Encompasses a wide range of scales: from single nucleotide polymorphisms (SNPs) through to structural variation involving millions of base pairs
- **Diversity:** A summary of variation at group level
 - Many measures: **heterozygosity**, nucleotide diversity, population mutation parameter theta (θ), proportioning of diversity within and between groups (F_{ST})...

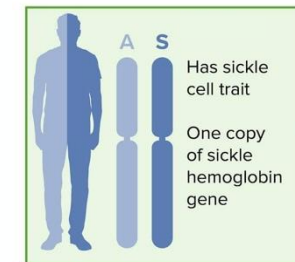


$F_{ST}=0$ (0% of the variation is apportioned between groups)

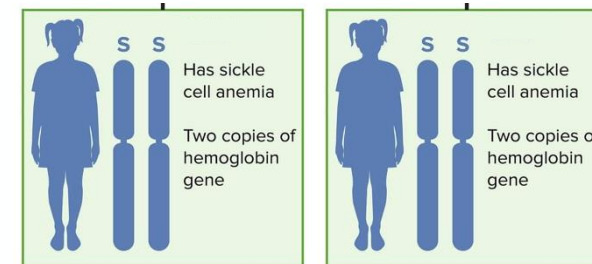
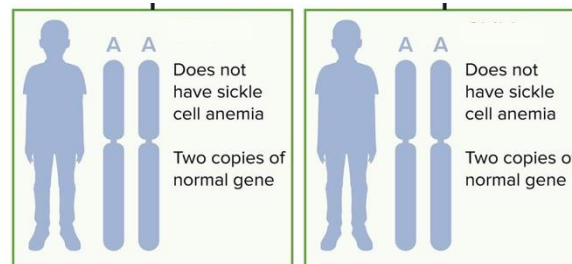


What is genetic diversity and **variation**, and **how do these arise?**

- **Variation:** The differences between genomes
 - Encompasses a wide range of scales: from single nucleotide polymorphisms (SNPs) through to structural variation involving millions of base pairs
- **Diversity:** A summary of variation at group level
 - Many measures: heterozygosity, nucleotide diversity, population mutation parameter theta (θ), proportioning of diversity within and between groups (F_{ST})...



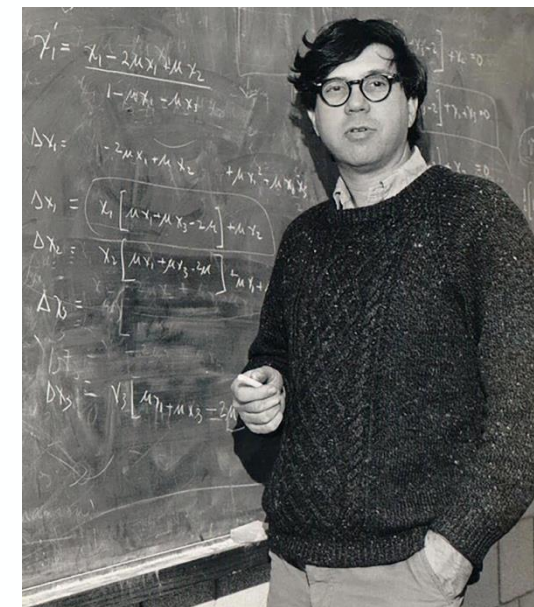
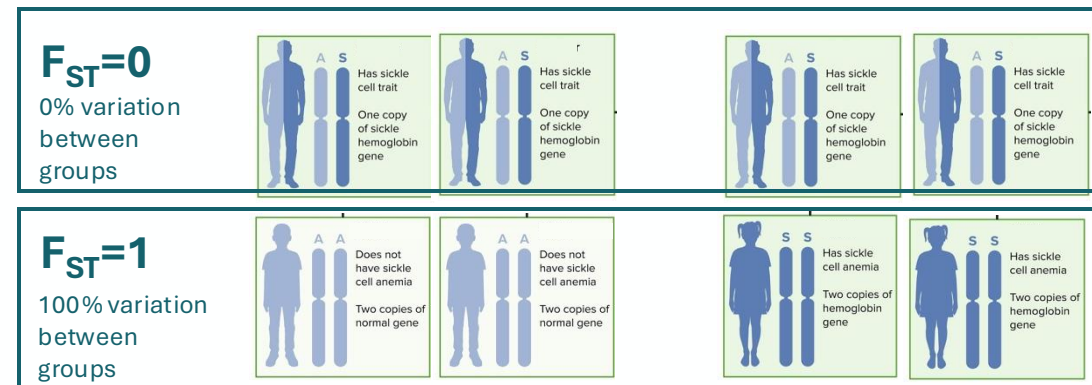
$F_{ST}=1$ (100% of the variation is apportioned between groups)



How much variation is there *between* groups versus *within* them?

Table 1. Human Genes or "Systems" Included in this Study and Extremes of Allele Frequency in Known Populations

Locus	Allele	Frequency Range	Extreme Populations
Haptoglobin	(Hp)	Hp ¹	Tamils-Lacondon
Lipoprotein	(Ag)	Ag ^x	Italy-India
Lipoprotein	(Lp)	Lp ^a	Labrador-Germany
	(Xm)	Xm ^a	Easter Is.-U.S. Blacks
Red Cell Acid Phosphatase	(APh)	p ^a	Tristan da Cunha-Athabascan
		p ^b	Athabascan-Tristan da Cunha
		p ^c	Many
6-phosphogluconate dehydrogenase	(6PGD)	PGD ^A	Bhutan-Yucatan
Phosphoglucomutase	(PGM ₁)	PGM ₁	Habbana Jews-Yanomama
Adenylate kinase	(AK)	AK ²	Africans, Amerinds-Pakistanis
Kidd	(Jk)	JK ^a	Chinese-Dyaks, Eskimo
Duffy	(Fy)	Fy ^a	Bantu-Chenchu, Eskimo
Lewis	(Le)	Le ^b	Lapps-Kapinga
Kell	(K)	K	Many-Chenchu
Lutheran	(Lu)	Lu ^a	Many-Brazilian Amerinds
P		P	Chinese-West Africans
MNS		MS	Oceanians-Bloods
		Ms	Papuans-Malays
		NS	Borneo, Eskimo-Chenchu
		Ns	Navaho-Palauans
Rh		CDe	Luo-Papuans
		Cde	Many-Chenchu
		cDE	Luo, Dyak-Japanese
		cdE	Many-Ainu
		cDe	Many-Luo
		cde	Many-Basques
ABO		I ^A	Toba-Bloods
		I ^B	Amerinds, Austr. Abo.-Toda
		i	Oraon-Toba



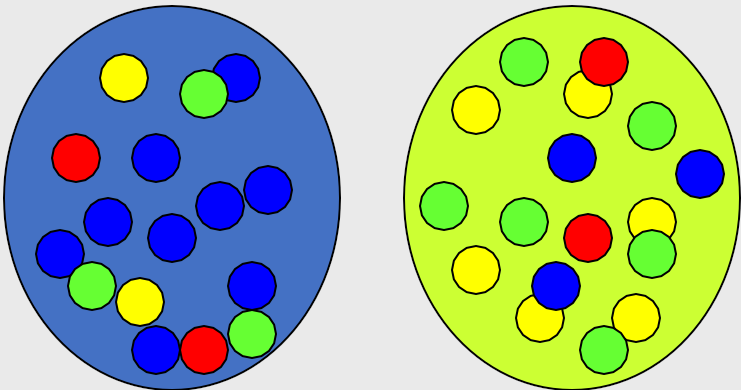
Lewontin (1972)

How much variation is there *between* groups versus *within* them?

Table 1. Human Genes or "Systems" Included in this Study and Extremes of Allele Frequency in Known Populations

Locus	Allele	Frequency Range	Extreme Populations
Haptoglobin	(Hp)	Hp ¹	Tamils-Lacondon
Lipoprotein	(Ag)	Ag ^x	Italy-India
Lipoprotein	(Lp)	Lp ^a	Labrador-Germany
	(Xm)	Xm ^a	Easter Is.-U.S. Blacks
Red Cell Acid Phosphatase	(APh)	p ^a	Tristan da Cunha-Athabascan
		p ^b	Athabascan-Tristan da Cunha
		p ^c	Many
6-phosphogluconate dehydrogenase	(G6PD)	PGDA	Phylos. Yucatan
Phosphoglucose isomerase	(GPI)	PGDA	Phylos. Yucatan
Adenylate kinase	(AK)	PGDA	Phylos. Yucatan
Kidd	(K)	PGDA	Phylos. Yucatan
Duffy	(D)	PGDA	Phylos. Yucatan
Lewis	(L)	PGDA	Phylos. Yucatan
Kell	(K)	PGDA	Phylos. Yucatan
Lutheran	(L)	PGDA	Phylos. Yucatan
P	(P)	PGDA	Phylos. Yucatan
MNS	(MNS)	PGDA	Phylos. Yucatan
Rh	(Rh)	PGDA	Phylos. Yucatan
ABO	(ABO)	PGDA	Phylos. Yucatan

Example of one gene with 4 different variants (alleles) in populations A and B

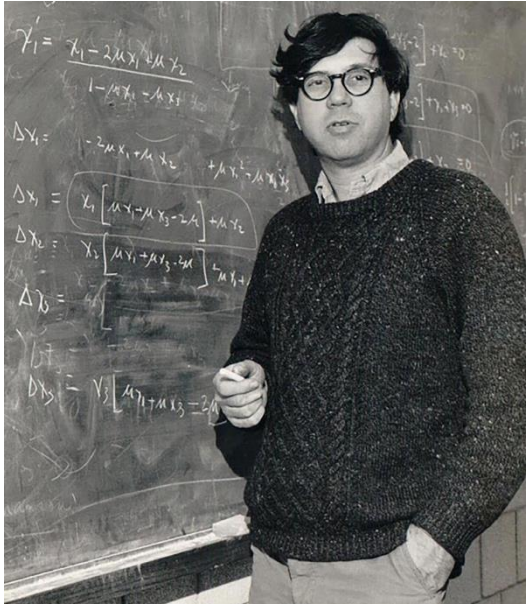
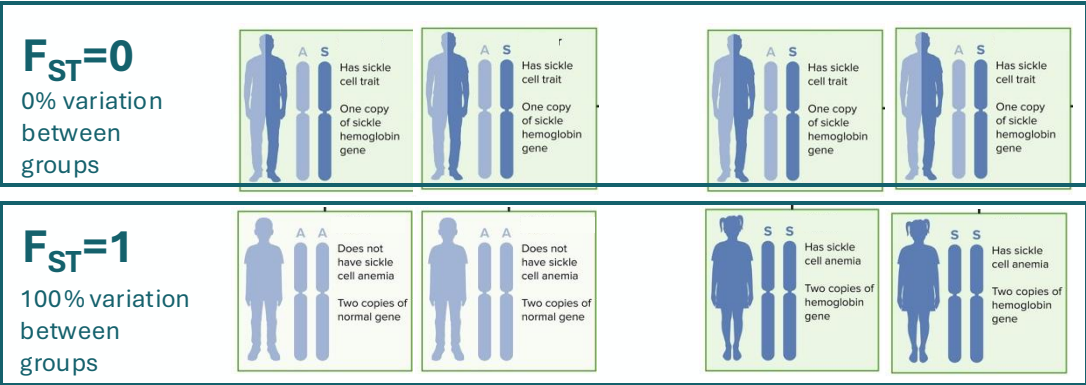


Handwritten: *At random loci then averaged*

Key

Human F_{ST} is 0.15

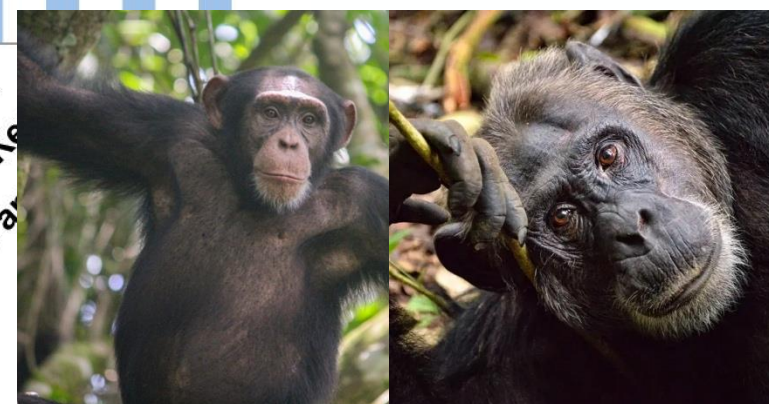
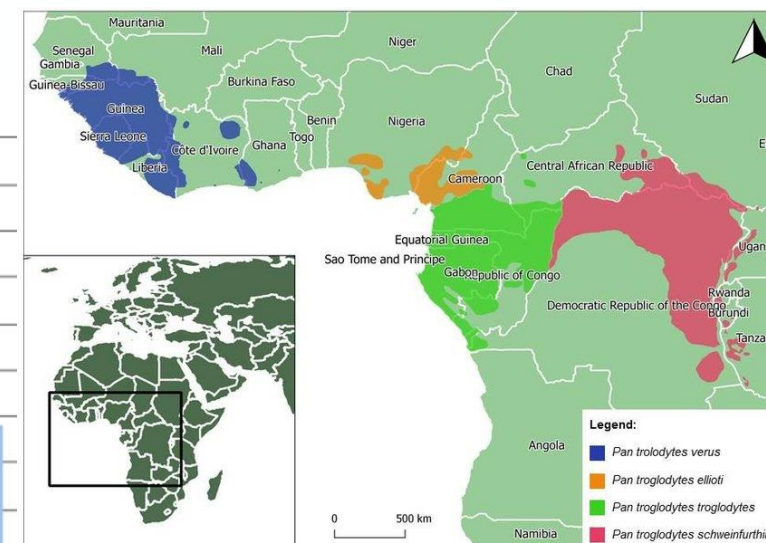
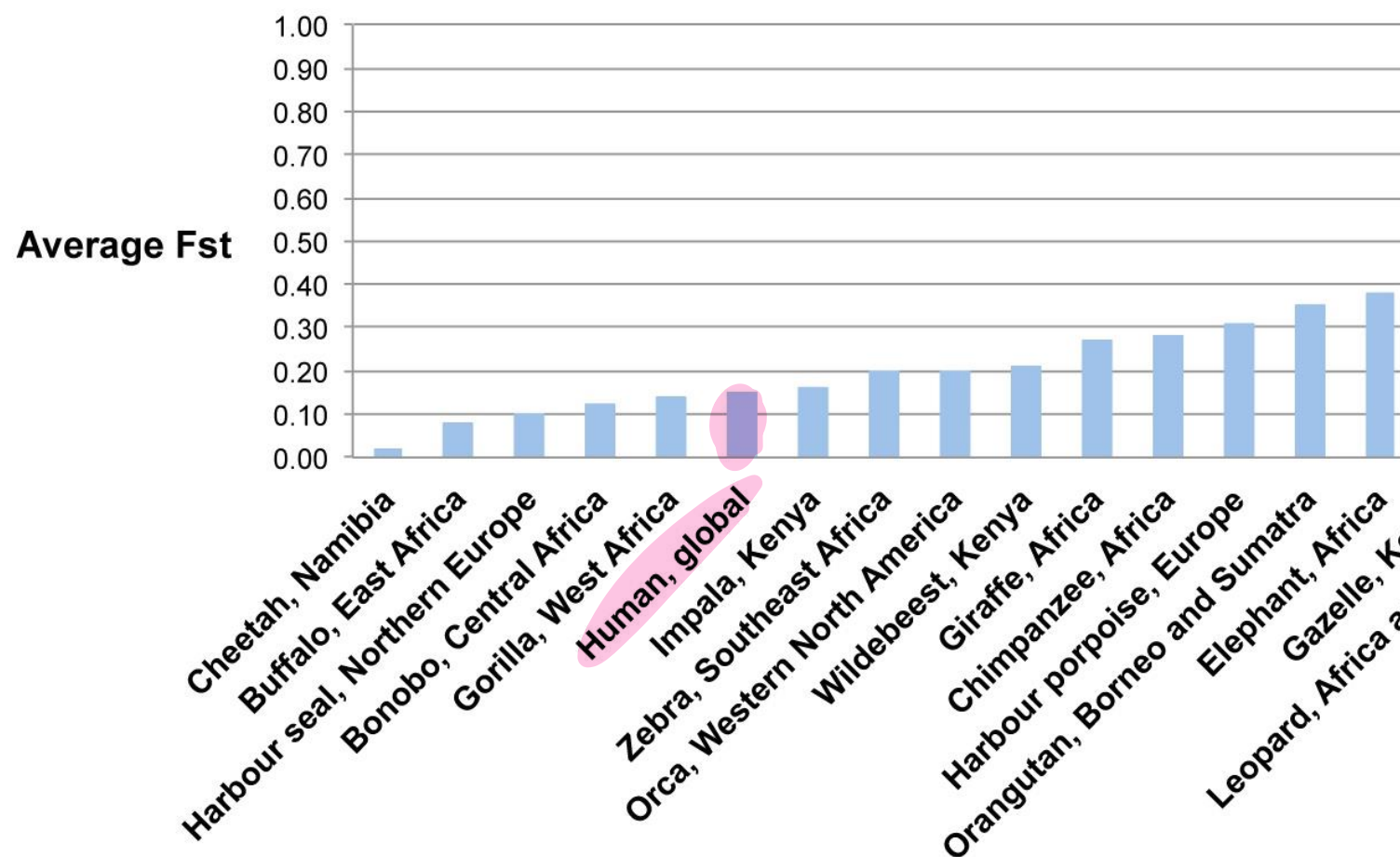
so 85% of variation is *within* populations



Lewontin (1972)

Surprisingly low F_{ST} for an ape, for a globally distributed mammal, etc

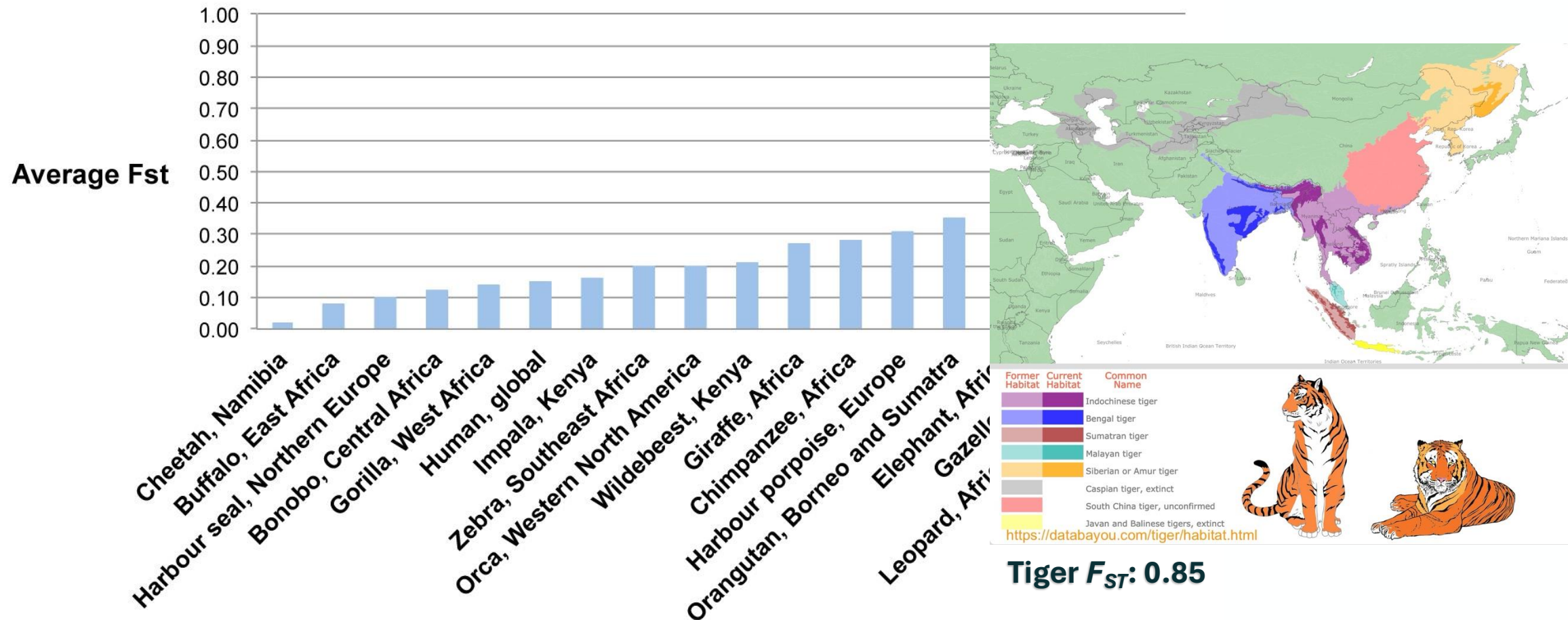
Key



Chimpanzee F_{ST} : 0.32

Surprisingly low F_{ST} for an ape, for a globally distributed mammal, etc

Key



Seven main takeaways

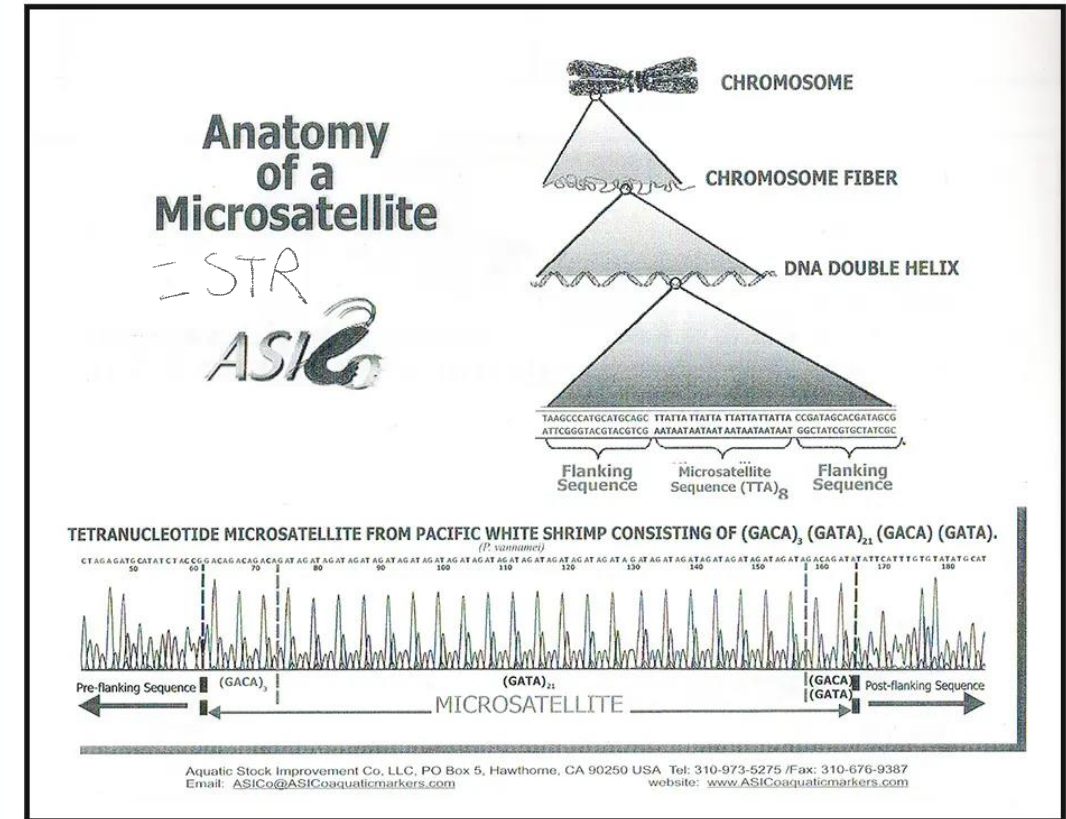
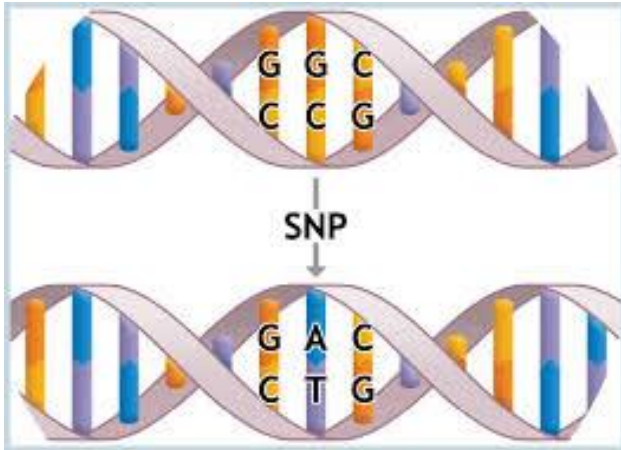
Patterns (what?)

1. Many human traits are nonconcordant
2. Most human genetic variation is clinal and/or continuous (not discrete)
3. **Genetic variation is greater within groups than between groups**
4. **Humans have (surprisingly?) high genetic homogeneity**

Processes (why?)

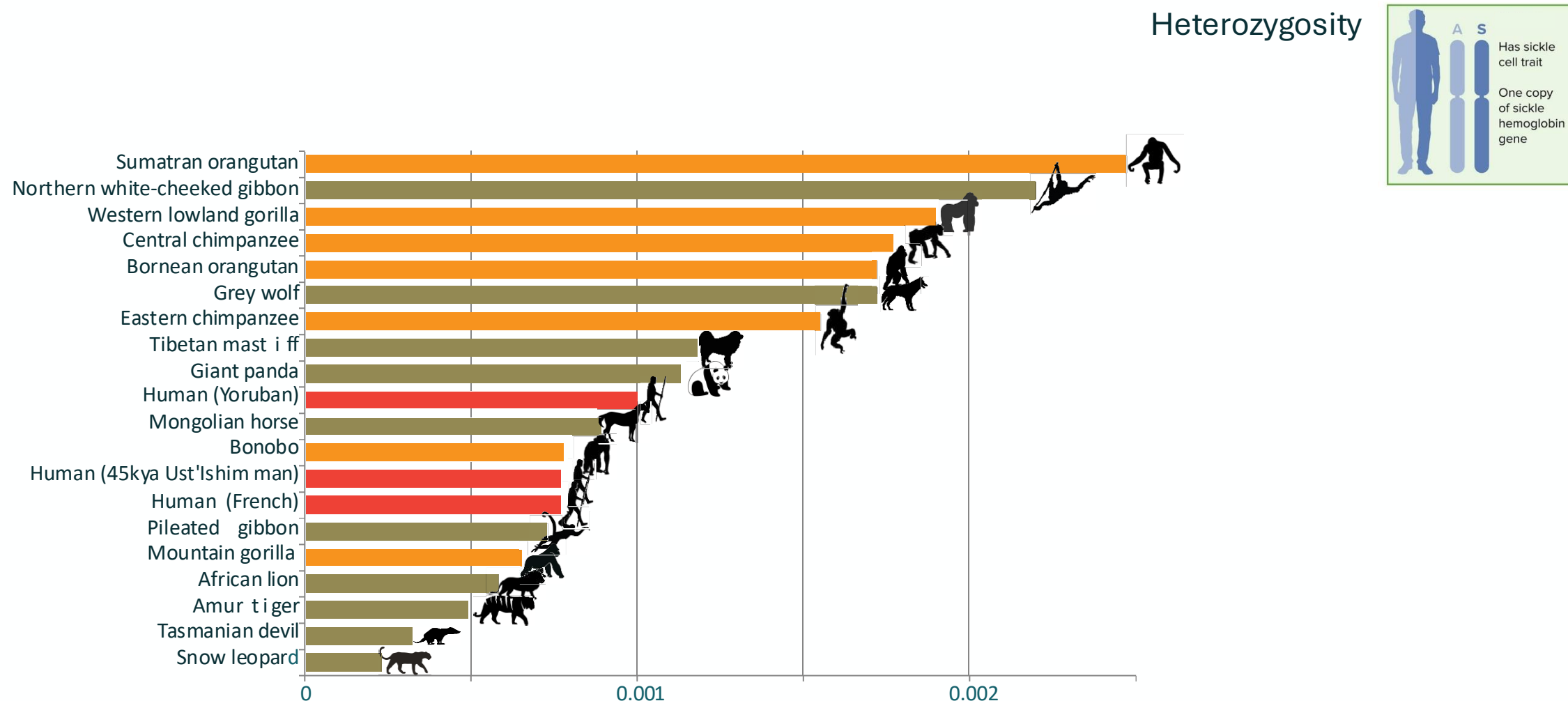
1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit neutral models
3. ...and others are explained by natural selection

SNPs and microsatellites



More variation across
genotypes

Humans have extraordinarily low genetic diversity



Average genome-wide heterozygosity in humans (red bars) is much lower than in other great apes (orange bars) and most mammals.

Humans have extraordinarily low genetic diversity

... *why?*

Low human diversity is fundamentally due to low ancient population size.

- When did that happen? Long-term or recent?
- To whom? All people (e.g. speciation bottleneck) or some (e.g. Out of Africa)?



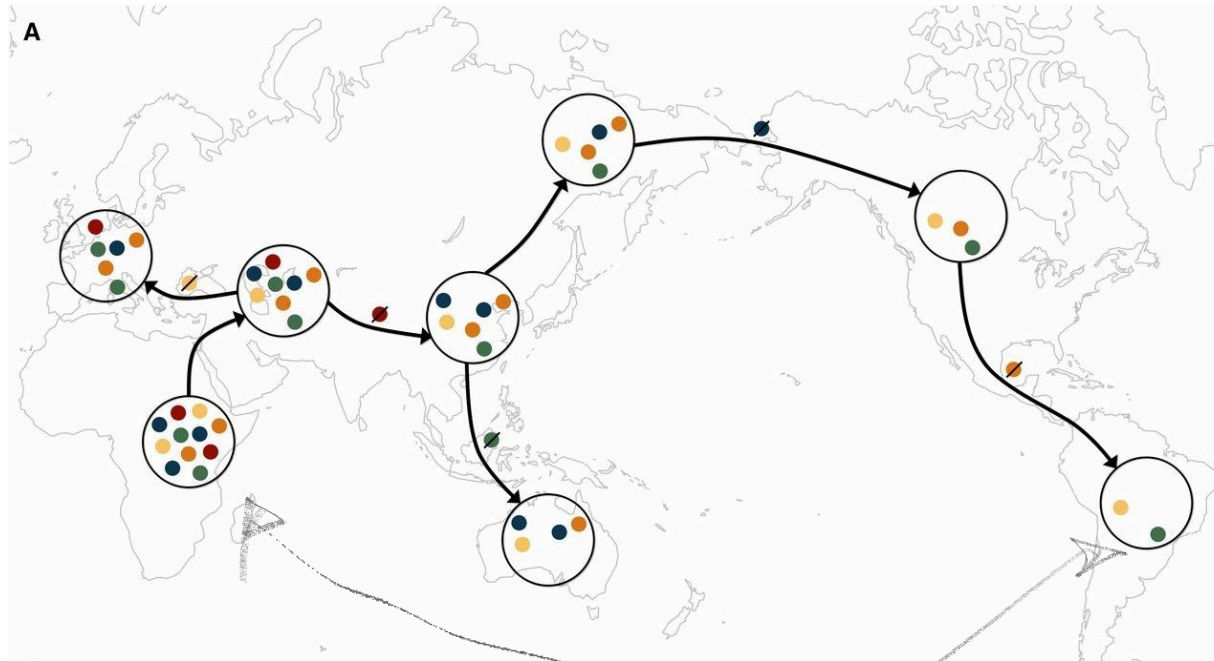
Expect diversity to be low
across the species
compared to other species



1. Expect diversity to be even lower outside of Africa
2. Expect diversity outside of Africa to be a subset of diversity within Africa

Humans have extraordinarily low genetic diversity

... why?



Serial founder effects

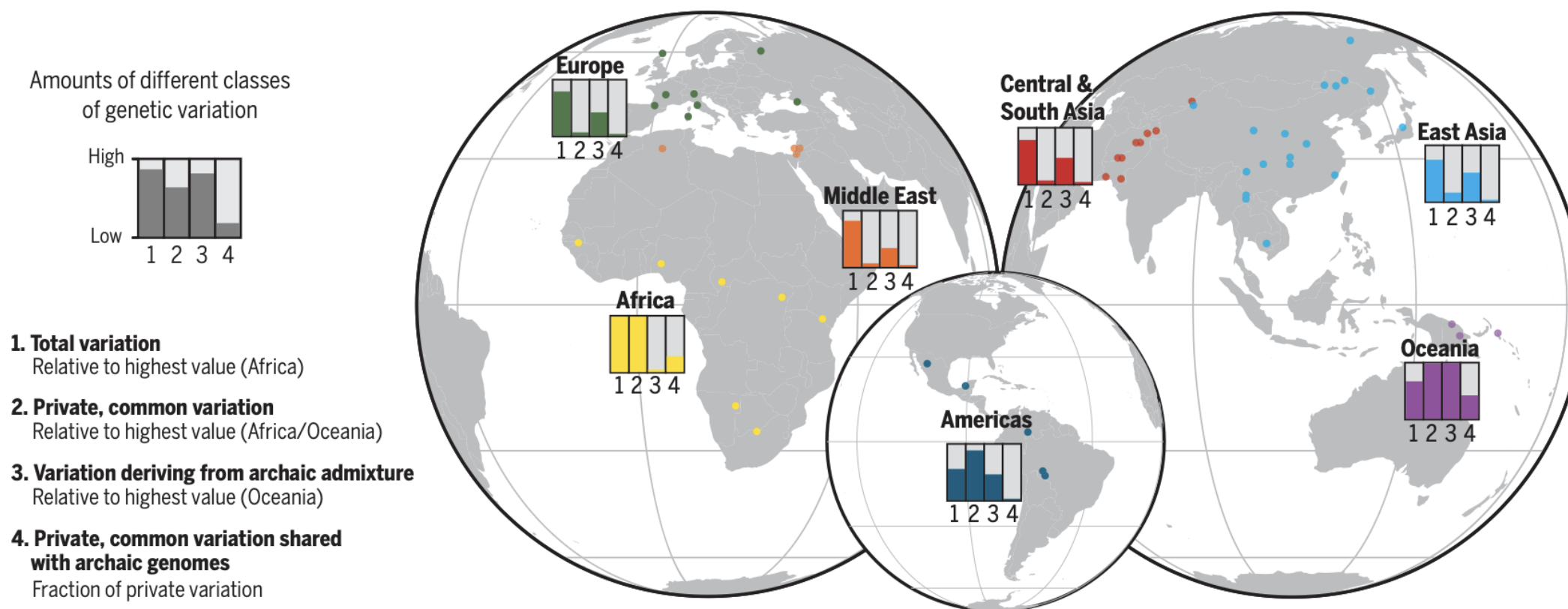
variation in bottleneck species
is still contained in some pop

Out of Africa



1. Expect diversity to be even lower outside of Africa
2. Expect diversity outside of Africa to be a subset of diversity within Africa

Diversity patterns across the whole genome



Non-African diversity is smaller than *and* a subset of African diversity

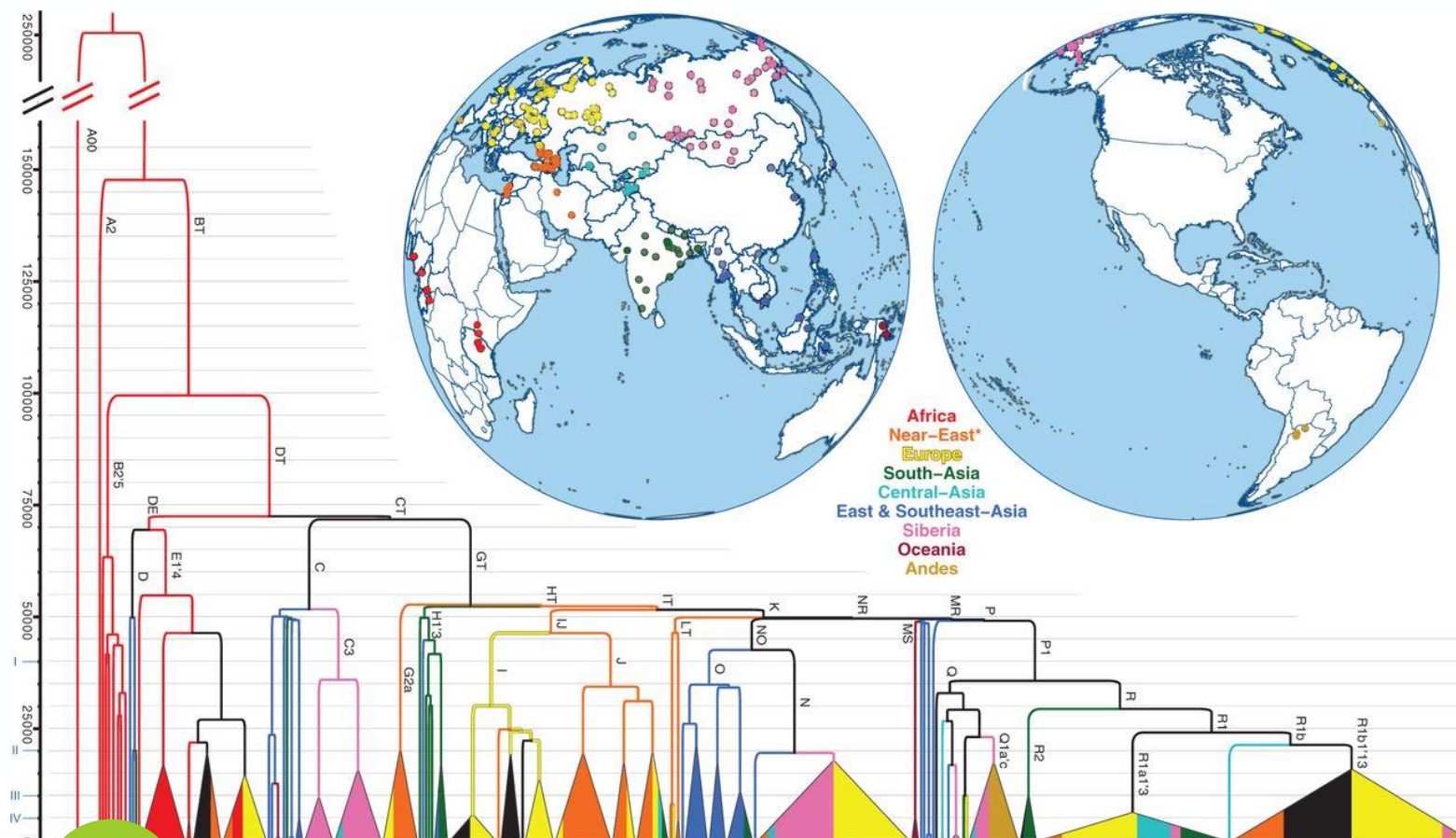


- Most genetic variants that are **common** are shared between continents
- Those that are region-specific are **rare**
- Highest genetic diversity in Africa
- Variation elsewhere subset of African variation

Bergström et al. (2020)

Y-chromosome & mtDNA tell the same story

The Y chromosome haplotypes of all non-African men derive from a single African male founder haplotype



Non-African diversity is smaller than *and* a subset of African diversity



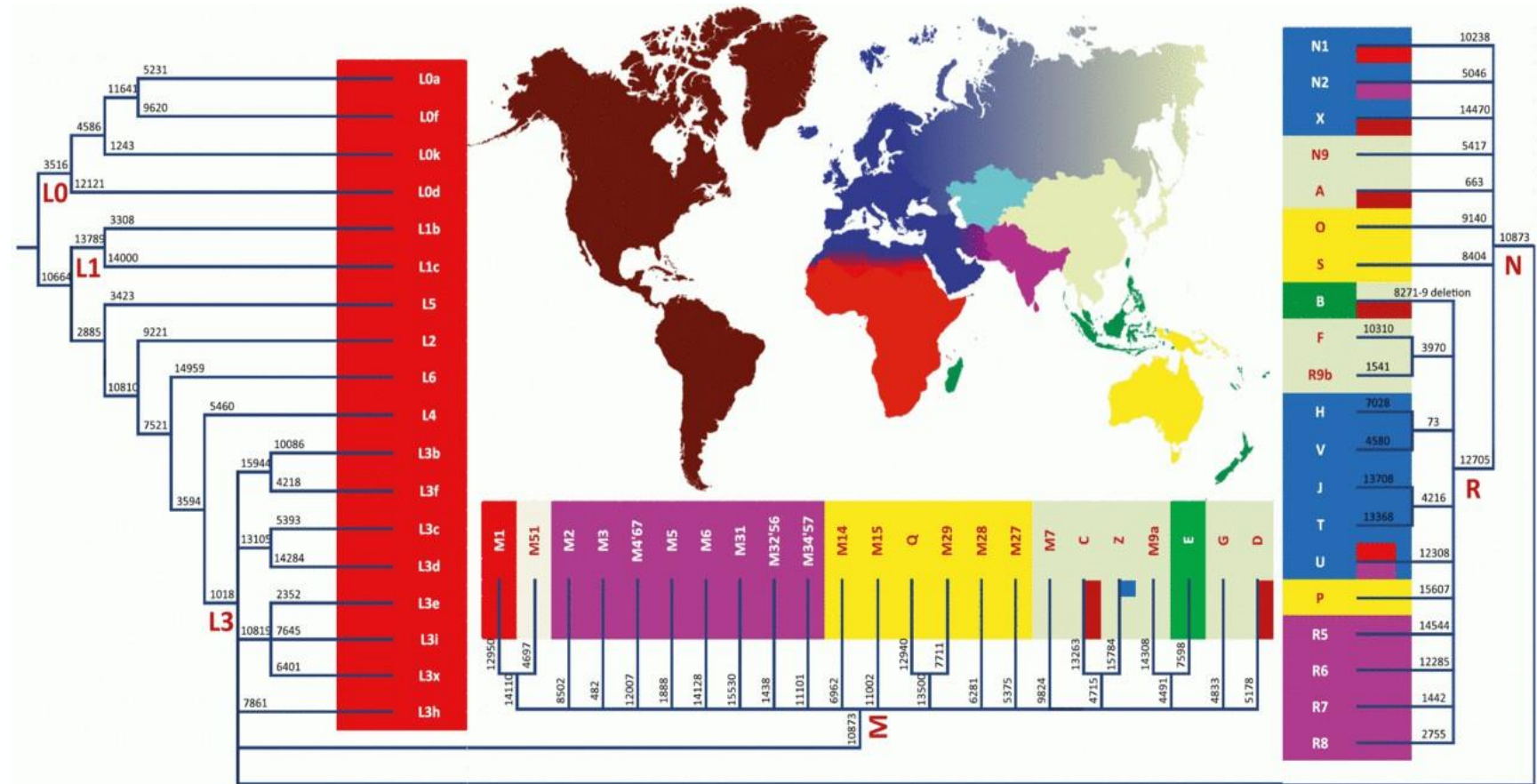
Lower SNPs downstream
of root SNPs

Karmin et al. (2015)

- Highest genetic diversity in Africa
- Variation elsewhere subset of African variation

Y-chromosome & mtDNA tell the same story

The mtDNA haplogroups of all non-African people derive from the L3 lineage in Africa



Kivisild (2005)

Non-African diversity is  smaller than *and* a subset of African diversity

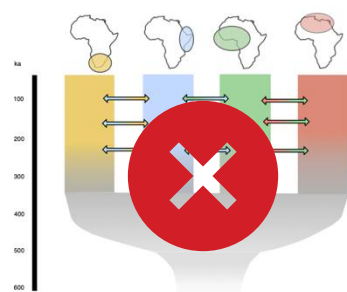
- Highest genetic diversity in Africa
- Variation elsewhere subset of African variation

Key

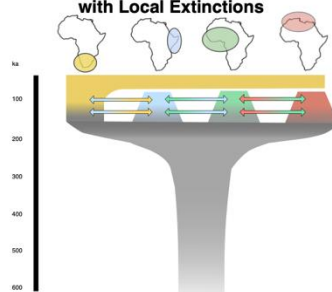
Humans have extraordinarily low genetic diversity

... why?

(a) Model 1: African Multiregionalism

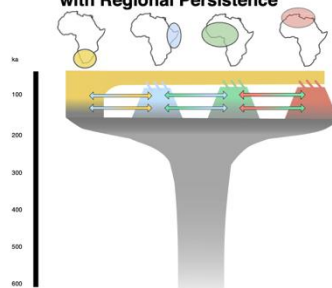


(b) Model 2: Single Origin Range Expansion with Local Extinctions



causes morphological diversity
and then founder group effect
either or will be true but requires modeling
s fundamentally due to low ancient population size.

(c) Model 3: Single Origin Range Expansion with Regional Persistence



Current Opinion in Genetics & Development



Expect diversity to be low
across the species
compared to other species

1? Long-term or recent?
(e.g. speciation bottleneck) or some (e.g. Out of Africa)?



1. Expect diversity to be
even lower outside of Africa
2. Expect diversity outside
of Africa to be a subset of
diversity within Africa

Key

Humans have extraordinarily low genetic diversity

... *why?*

To what extent are humans genetically locally adapted?

Low human diversity is fundamentally due to low ancient population size.

BUT does this mean we have no genetic adaptations? We'll see

- When did that happen? Long-term or recent?
- To whom? All people (e.g. speciation bottleneck) or some (e.g. Out of Africa)?



Expect diversity to be low across the species compared to other species



1. Expect diversity to be even lower outside of Africa
2. Expect diversity outside of Africa to be a subset of diversity within Africa

Seven main takeaways

Patterns (what?)

1. Many human traits are nonconcordant
2. Most human genetic variation is clinal and/or continuous (not discrete)
3. Genetic variation is greater within groups than between groups
4. Humans have (surprisingly?) high genetic homogeneity

Processes (why?)

1. Genetic data generally fit isolation-by-distance, serial bottlenecks, and Out-of-Africa models
2. Some genetic patterns fit neutral models
3. ...and others are explained by natural selection

Next lecture:

To what extent are humans genetically locally adapted?